

Life and Health Sciences

Course Description

- LHS provides the non-science major a strong and diverse background necessary to **understand the structural and functional diversity of organisms while providing students with basic scientific and biological principles.**
- **Topics include basic and biological chemistry, cell structure and processes, and DNA and genetic basics.** This course partially fulfills the undergraduate graduation requirements of HITSz. It may not be credited toward a biology degree.
- **It is important for all of us to have a good working knowledge of life, biology, health for day-to-day life.**

Increasingly, biology-related topics are appearing in newspapers, magazines, on SNS, and in politics.

Over the course of your life you will make important decisions regarding LHS issues, such as voting for politicians with specific stances; and influencing your community to take a stance on various issues and policies:

Should we produce and eat genetically modified organisms?

Should we allow some amount of stem cell use or cloning?

Further, the decisions you make for your own life, such as those that are health and medicine related, are all based on an underlying understanding of LHS.

- **LHS affects every aspect of our daily lives whether we are immediately aware of it or not.**

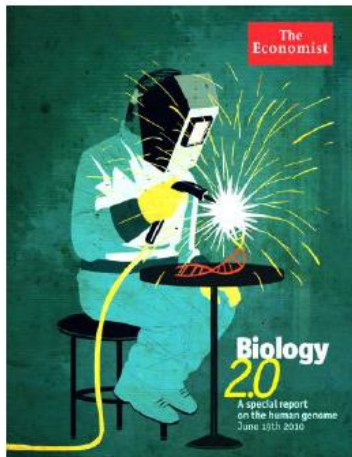
Life and Health Sciences -- Course Objectives

- 1) To examine the nature of science, the scientific method, & hypothesis testing
- 2) To examine cell diversity, structure, & function
- 3) To examine basic chemical principles, the nature of organic molecules, & the function of chemicals within cells
- 4) To examine the role of energy in maintaining life & learn how cells acquire & use energy
- 5) To examine the structure & function of DNA especially as it pertains to protein synthesis
- 6) To examine the principles of inheritance (genetics) & explore patterns of inheritance in humans
- 7) To examine the principles & regulation of cell division, & the consequences of malfunctions in the regulation of cell division (e.g. cancer)
- 8) To examine aspects of LHS & biotechnology, and discuss the role that biotechnology plays in our world, including an exploration of the ethics & consequences of emerging technologies
- 9) To examine the typical cases developed with LHS

Life and Health Sciences

Course Outcomes

- should show competency in each objective listed above.
- should be able to demonstrate understanding of basic principles of biology, have a conversational knowledge of the principles behind breaking news in LHS, and understand metabolism, physiology, and genetics sufficient to make wise decisions regarding health and nutrition.
- should also have an improved ability to think critically and to make informed decisions as a member of society.



Life and Health Sciences (LHS)

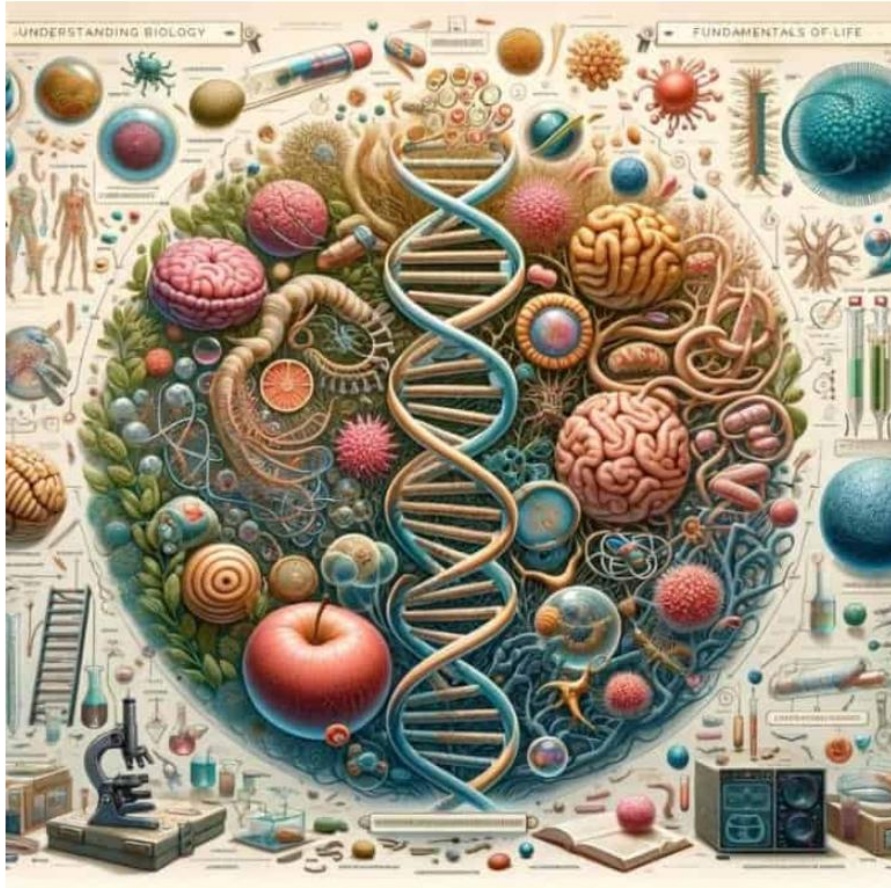
LHS covers the general knowledges of several fields in life sciences and health sciences!



You are not required to obtain comprehensive knowledge on all of these related fields ! Rather the general knowledge outlined in the objectives!

Importance of Life and Health Sciences

1. Understanding the Fundamentals of Life



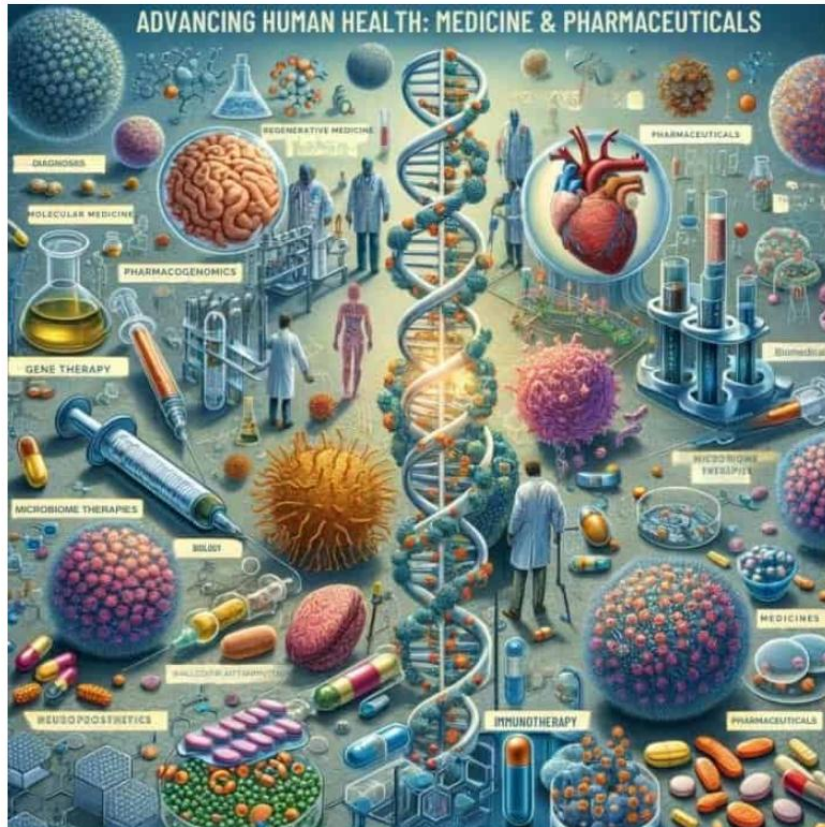
2. Unraveling Evolutionary Mysteries



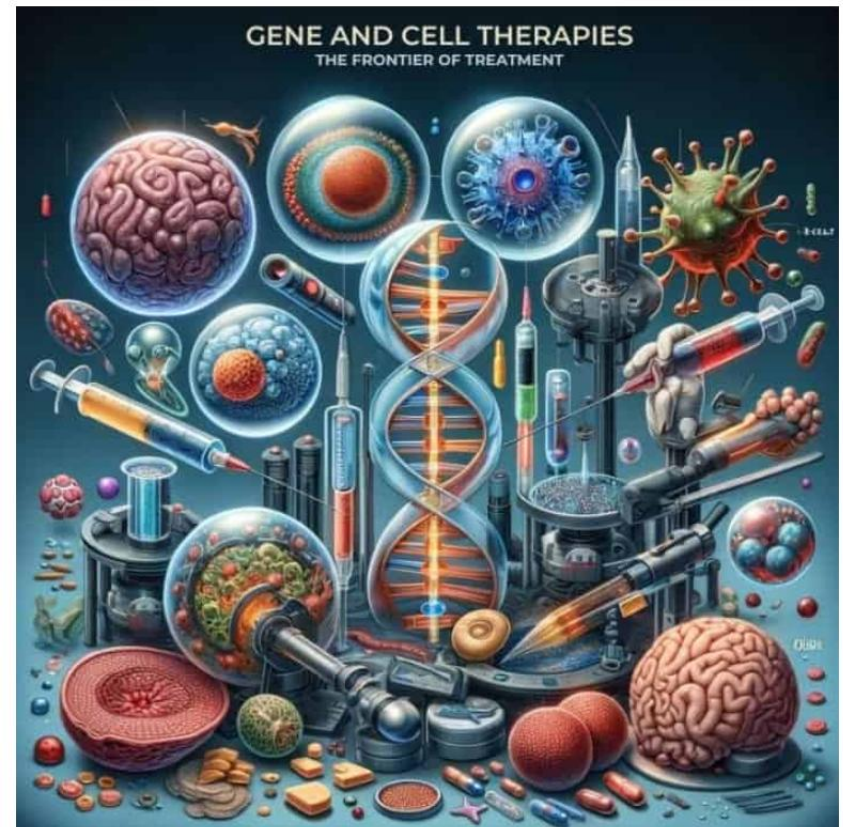
https://www.bioexplorer.net/importance-of-biology.html/#1_Understanding_the_Fundamentals_of_Life

Importance of Life and Health Sciences

3. Advancing Human Health: Medicine and Pharmaceuticals

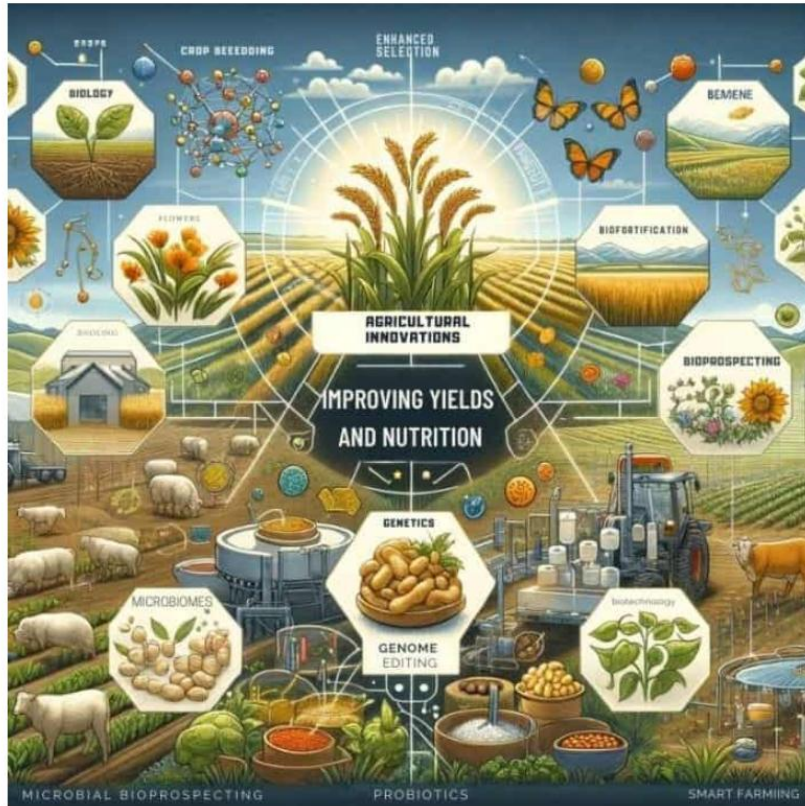


4. Gene and Cell Therapies: The Frontier of Treatment



Importance of Life and Health Sciences

5. Agricultural Innovations: Improving Yields and Nutrition

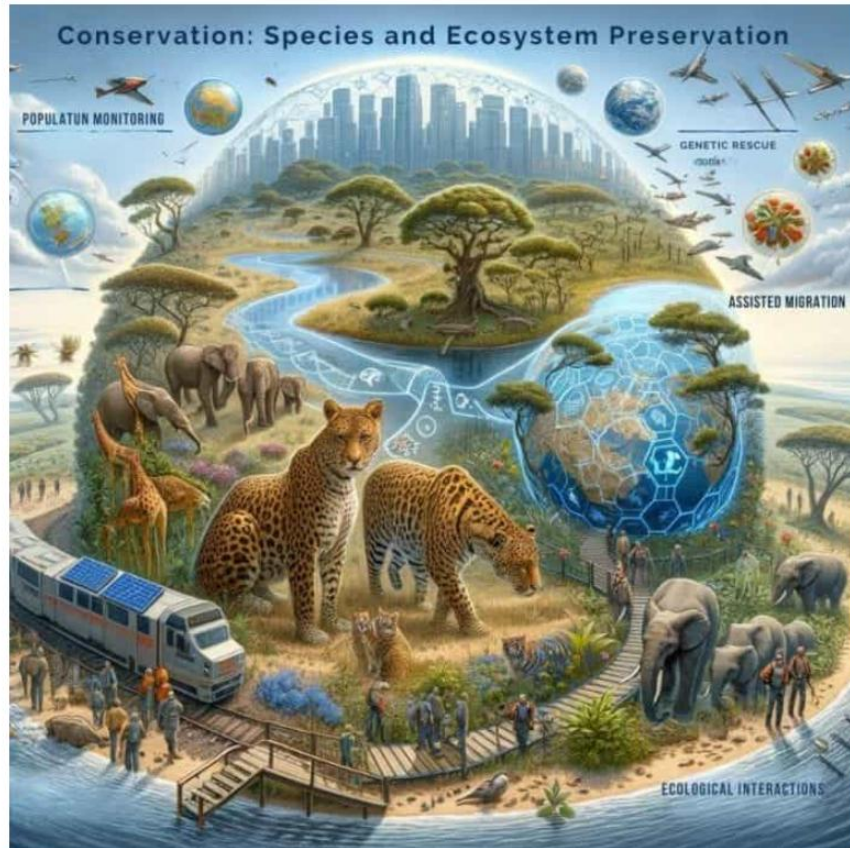


6. Sustainable Farming and Bioengineering



Importance of Life and Health Sciences

7. Conservation: Species and Ecosystem Preservation

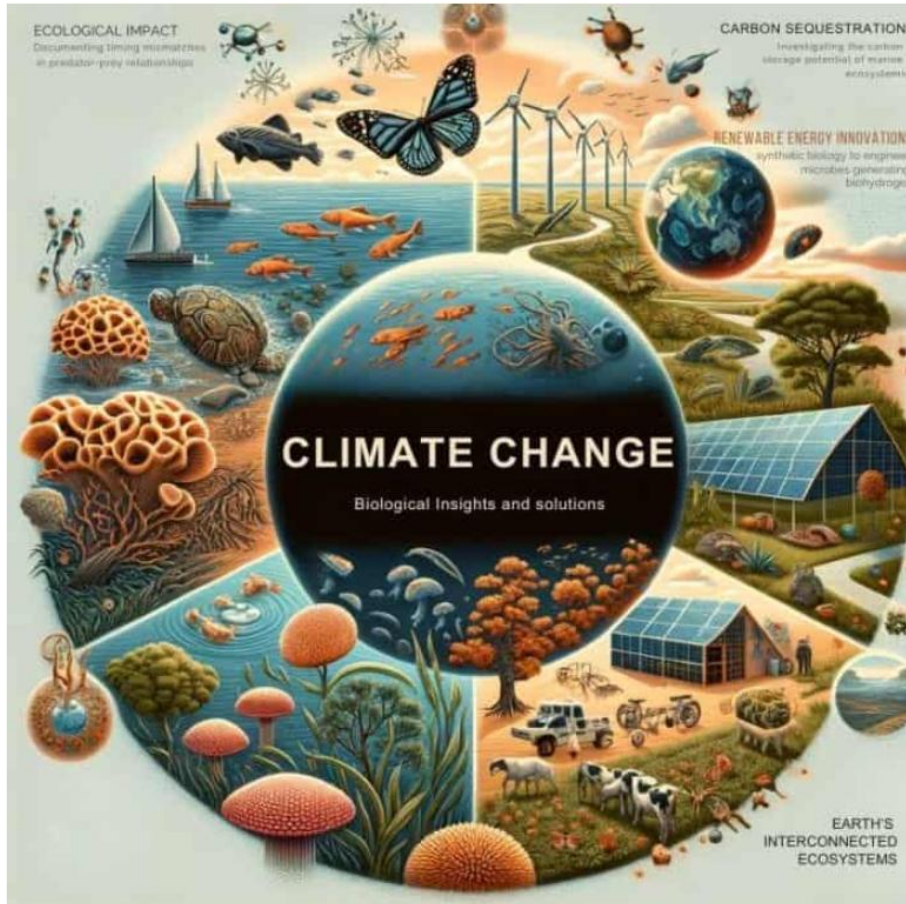


8. Habitat Restoration and Environmental Management

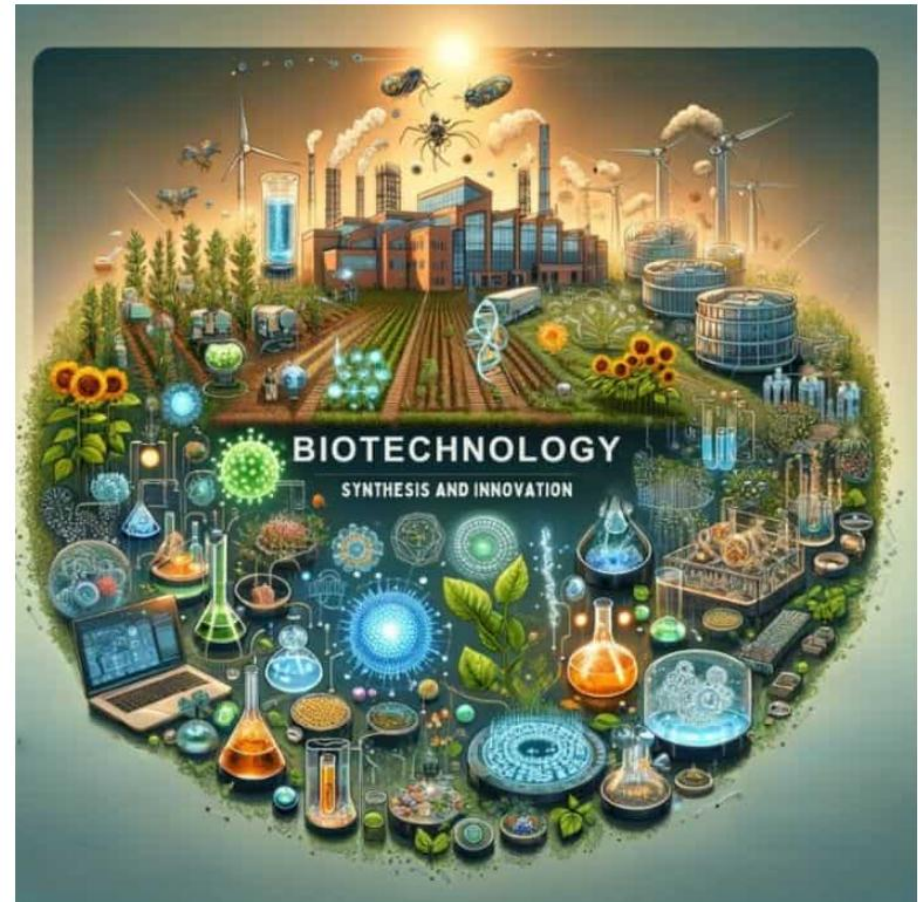


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9. Climate Change: Biological Insights and Solutions

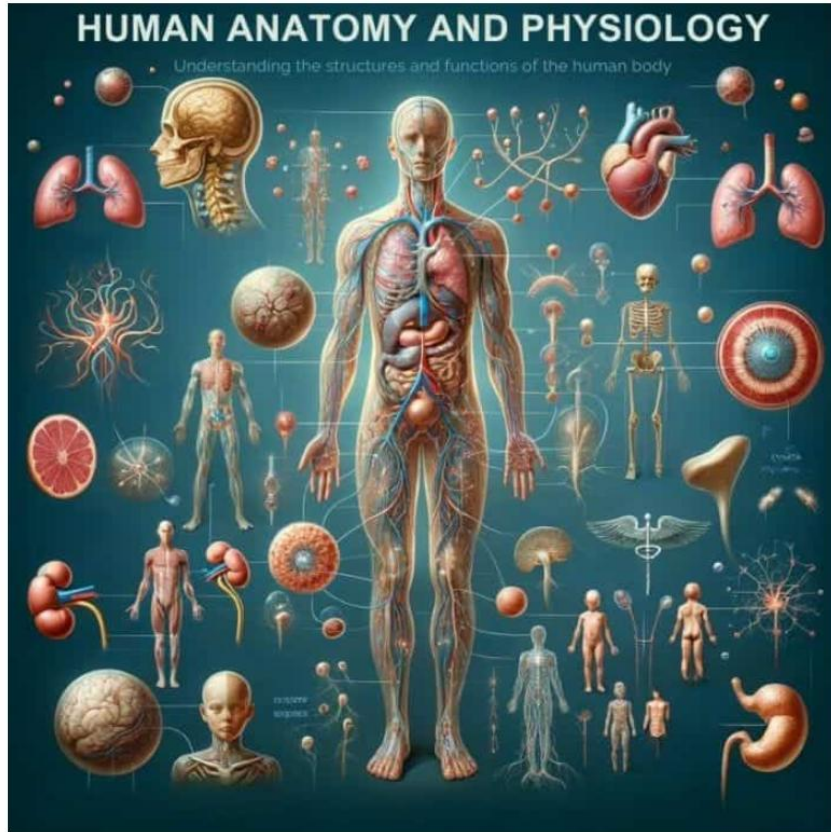


10. Biotechnology: Synthesis and Innovation

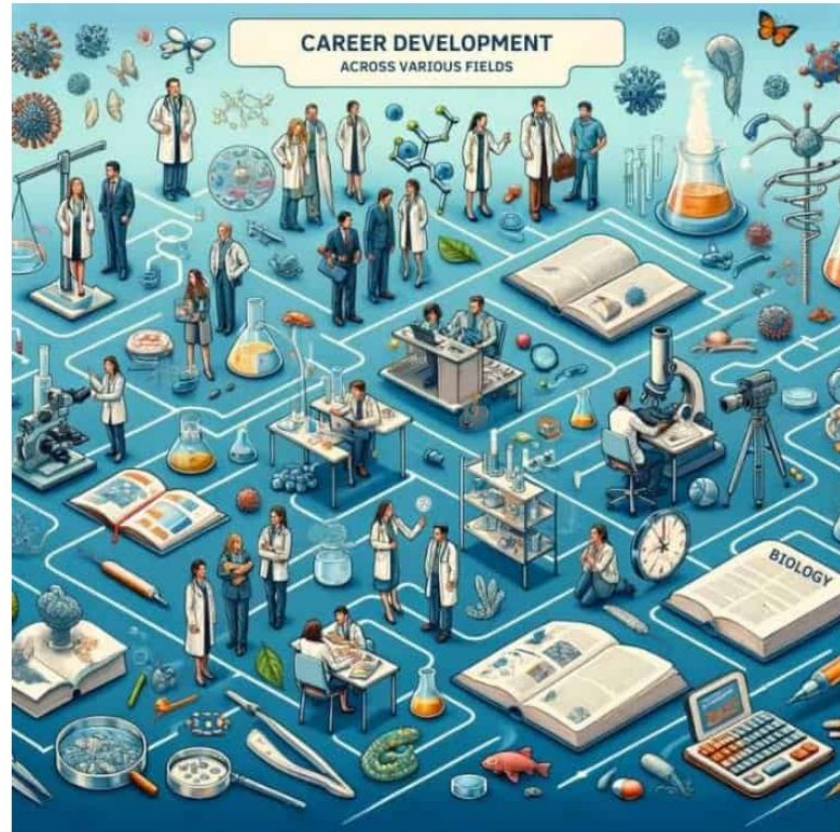


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11. Human Anatomy and Physiology

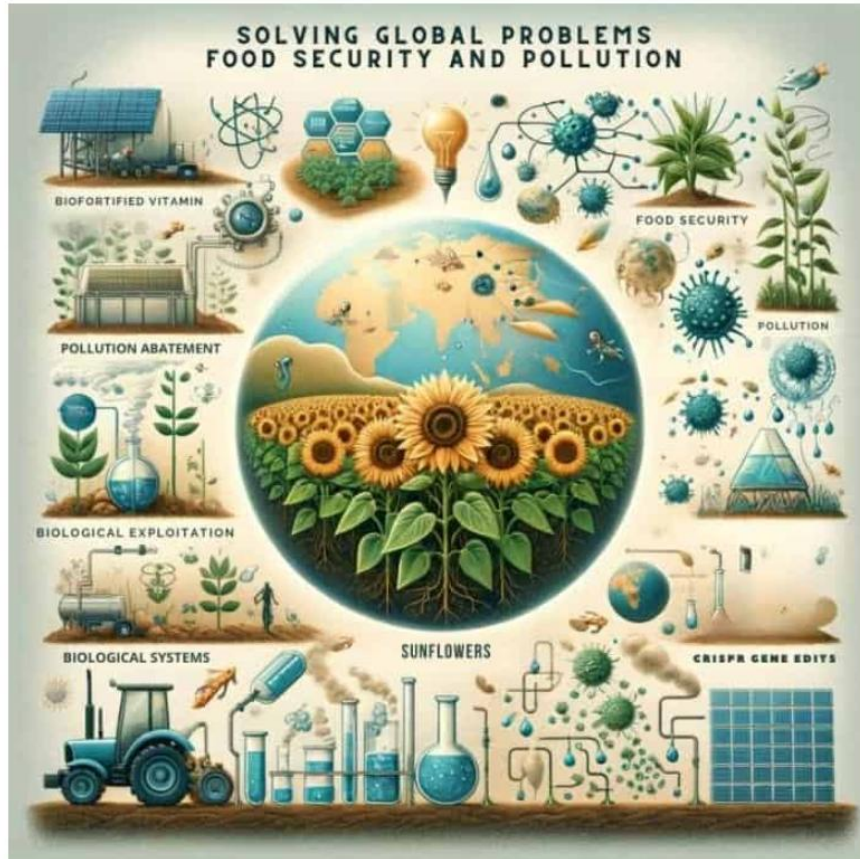


12. Career Development Across Various Fields



Importance of Life and Health Sciences

13. Solving Global Problems: Food Security and Pollution

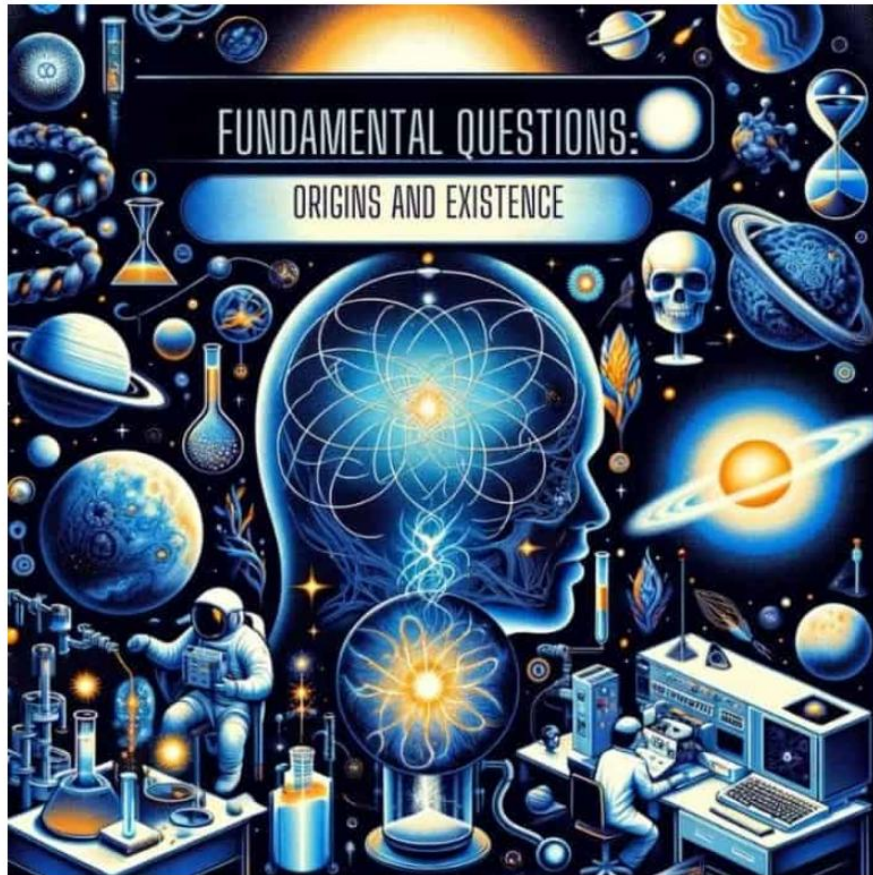


14. Basic Living Concepts and Lifestyle Applications

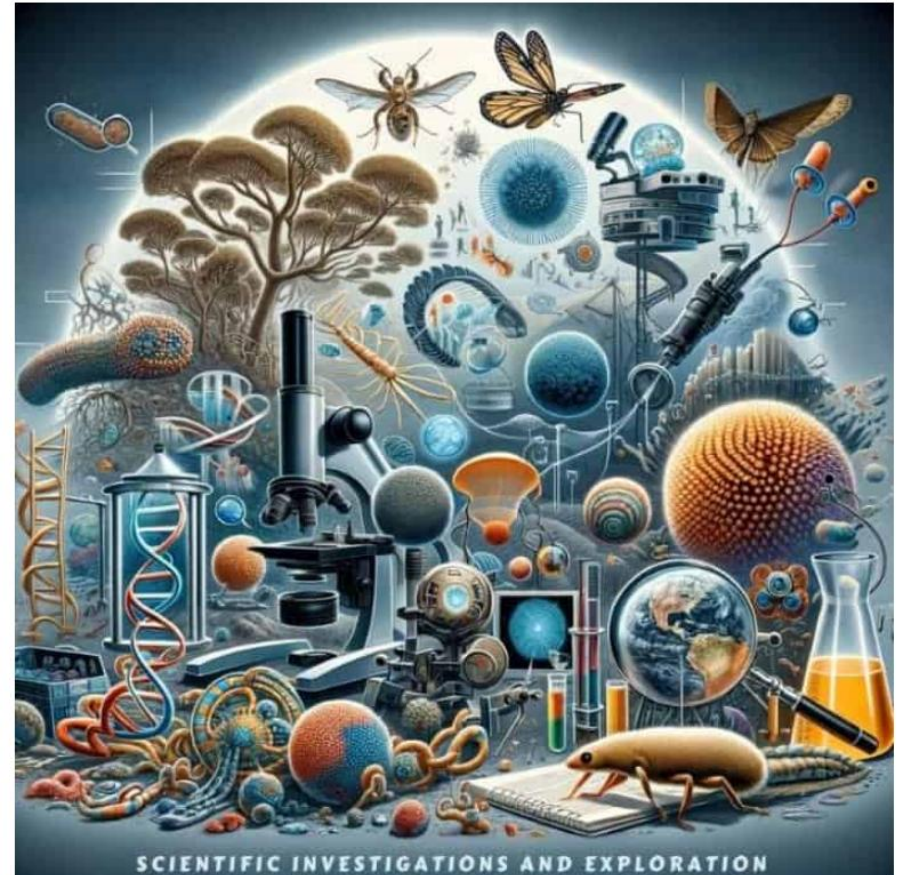


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15. Fundamental Questions: Origins and Existence

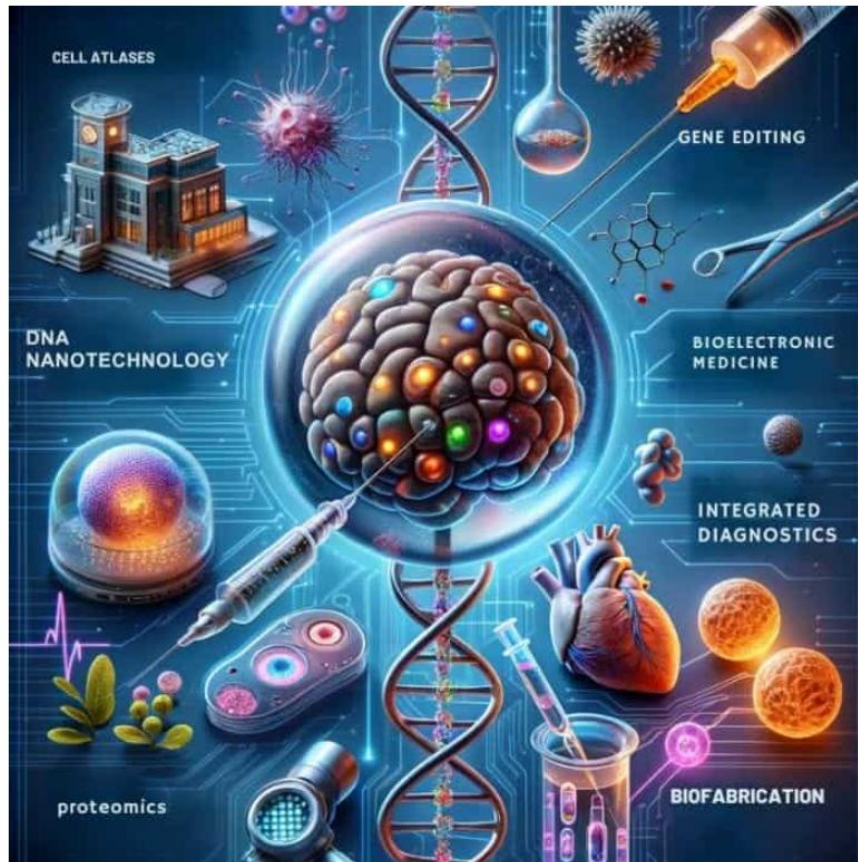


16. Scientific Investigations and Exploration

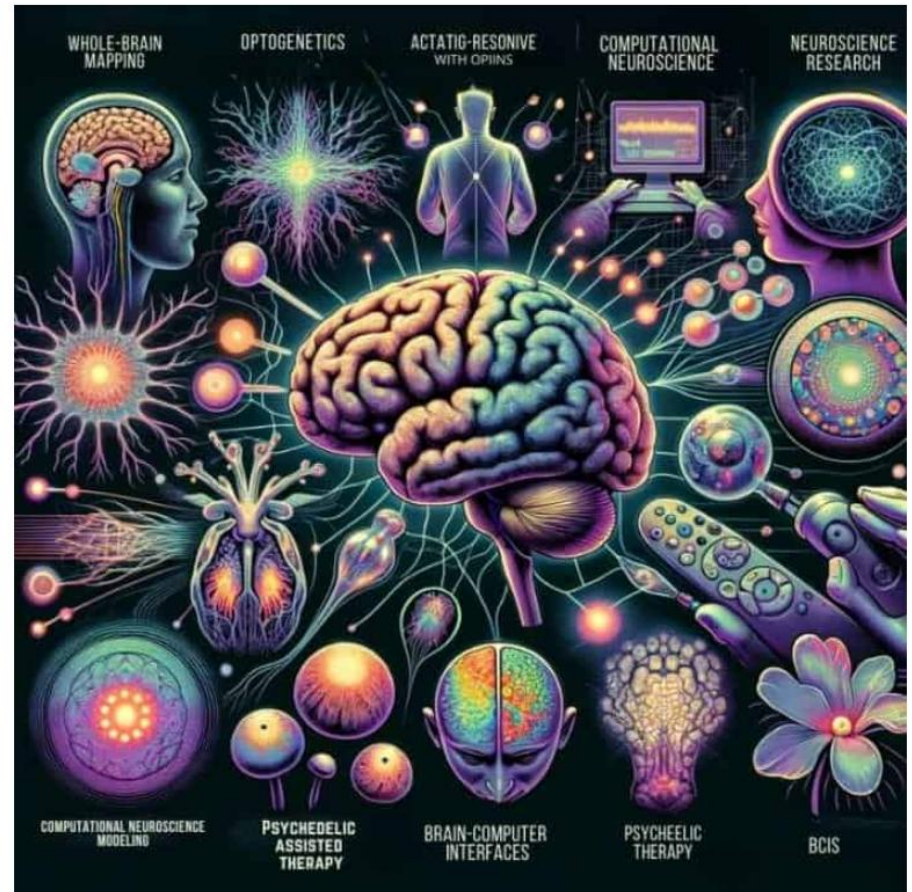


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17. Advanced Medical Research: Regenerative and Personalized Medicine



18. Neurobiology and Cognitive Sciences



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19. Botany and New Food Source Exploration



20. Bioinformatics and Computational Biology

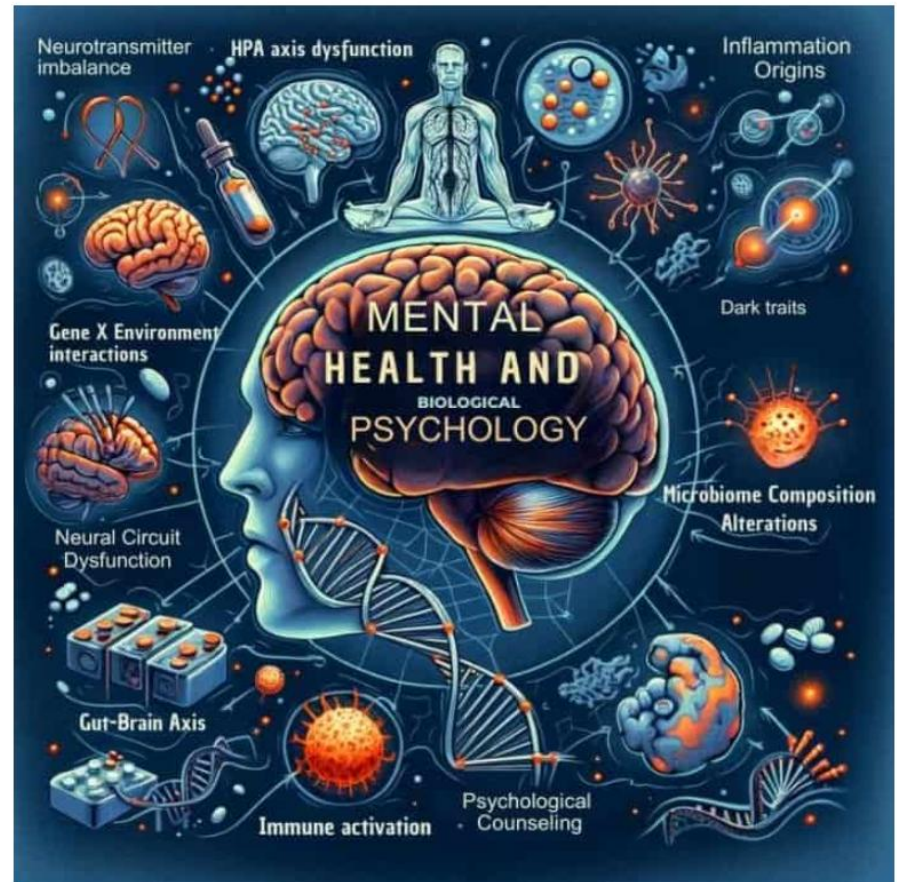


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21. Marine Biology and Ocean Ecosystems

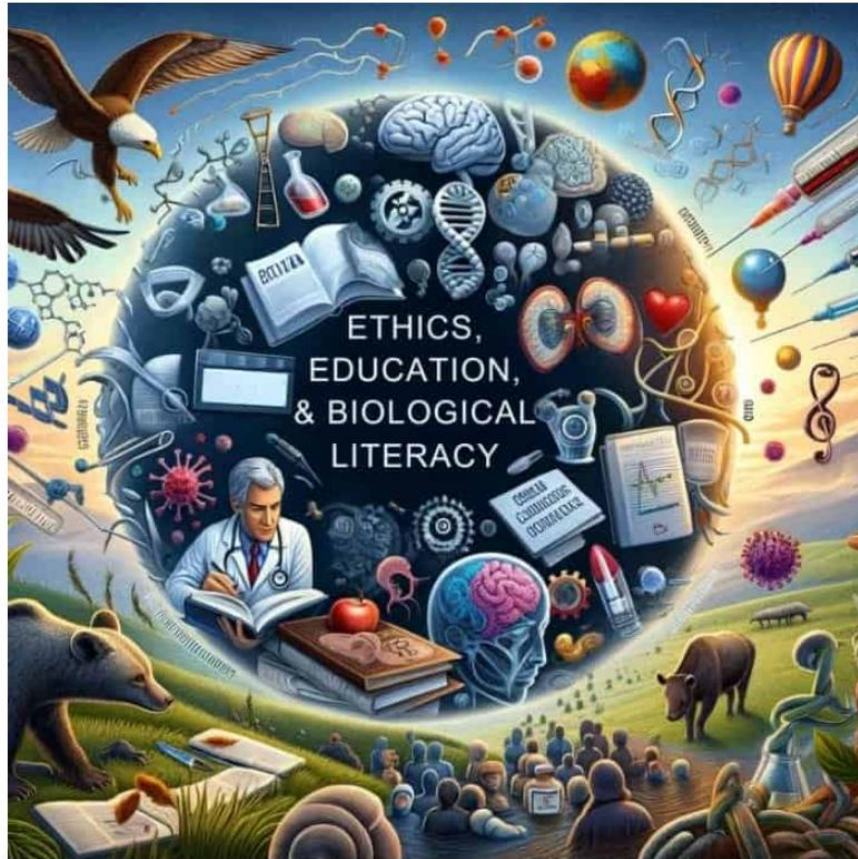


22. Mental Health and Biological Psychology

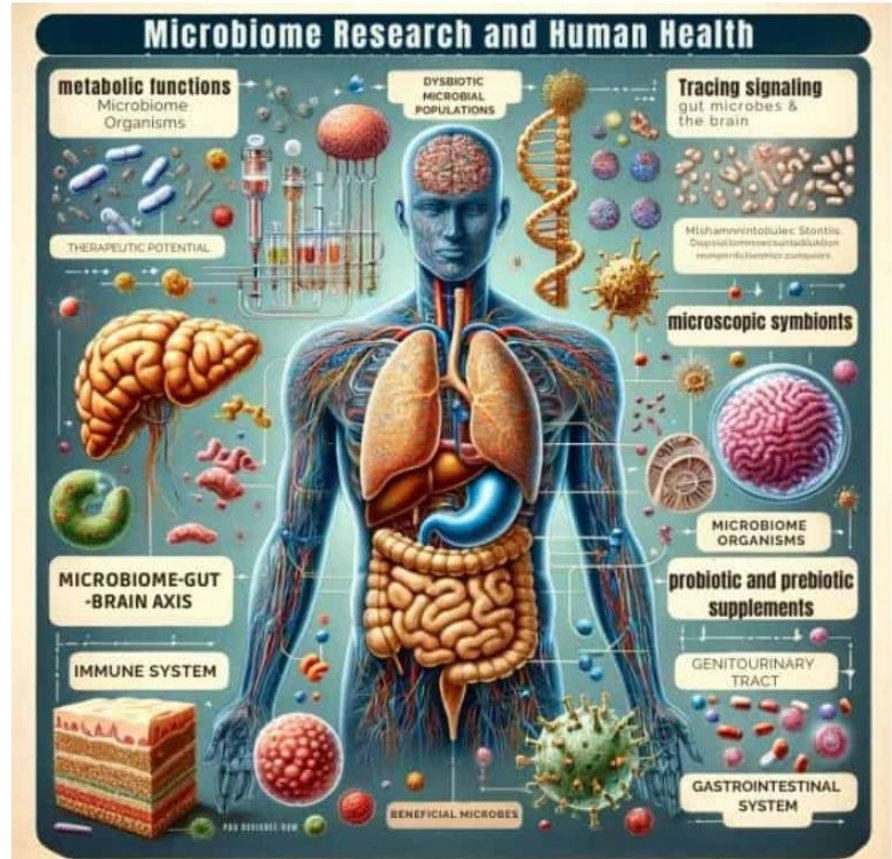


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23. Ethics, Education, and Biological Literacy

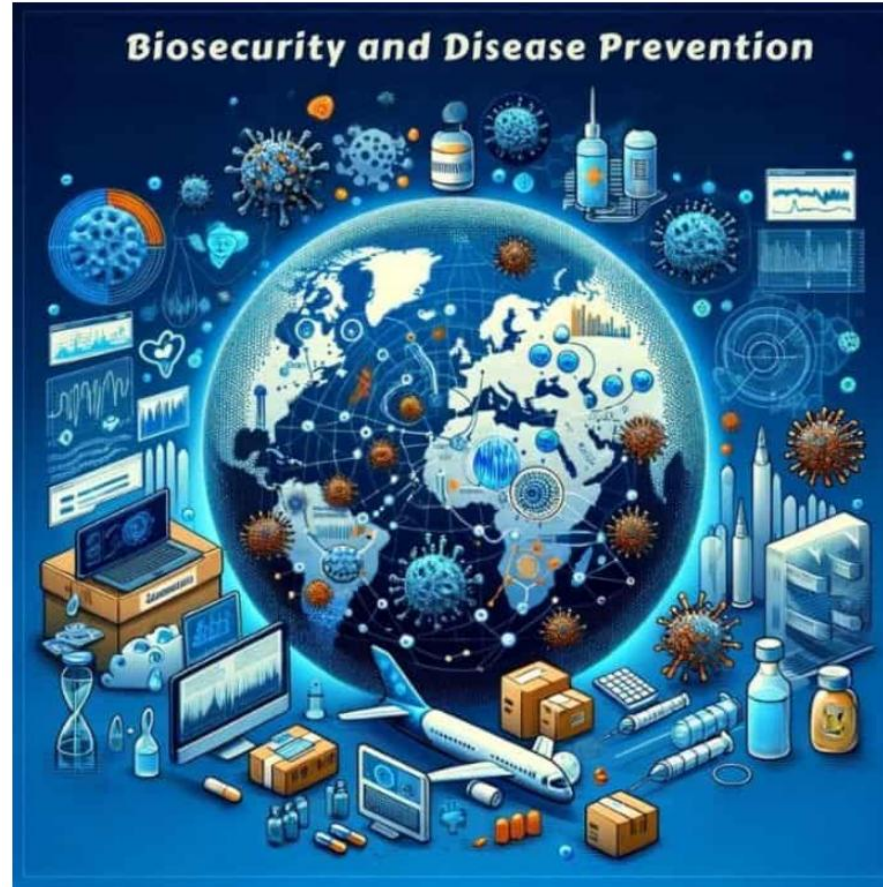


24. Microbiome Research and Human Health



Importance of Life and Health Sciences

25. Biosecurity and Disease Prevention



Importance of Life and Health Sciences

- These 25 points underscore the monumental societal impact of advancing humanity's understanding of life and living systems.
- From foundational basic research probing the fundamental workings of cells, organisms, and ecosystems to translational applications -- revolutionizing medicine, agriculture, conservation, and technology – biological sciences empower breakthrough innovations that tangibly transform lives.
- The ongoing investigation of genetics, physiology, evolution, symbioses, and more reveals emerging potentials limited only by our curiosity and resources.

Chapter 1: The Scientific Study of Life

- What is science & how do we “do” science?
- Why do scientists use the word “theory?”
- What is biology & what makes a thing “living?”

Corresponds with OpenStax Biology 2e Chapter 1

What is science & how do we “do” science?

- **Science** = an intellectual activity involving observation, description, experimentation, & explanation of the *natural world*
 - The natural world is the *objective & measurable* world around us
 - Science is NOT just a collection of facts, it's something you *do*
 - Science can NOT answer questions about immeasurable things
 - e.g. religion, morals, ethics, values, art, etc.

- Being a scientist is like being a detective
 - instead of answering questions about crimes, you answer questions about the natural world
- *Does the radiation from cell phones cause brain tumors?*
- *Does vitamin C reduce the likelihood of getting a cold?*
- *Does “blood doping” really improve athletic performance?*
- *Are indoor tanning salons safer than tanning outdoors?*
- *What dose of alcohol is lethal for a human?*
- *Which parent determines a baby’s sex?*
- *Does aspirin help with fevers?*
- *Do probiotics actually improve gut → health?*

photo by Mike Mozart



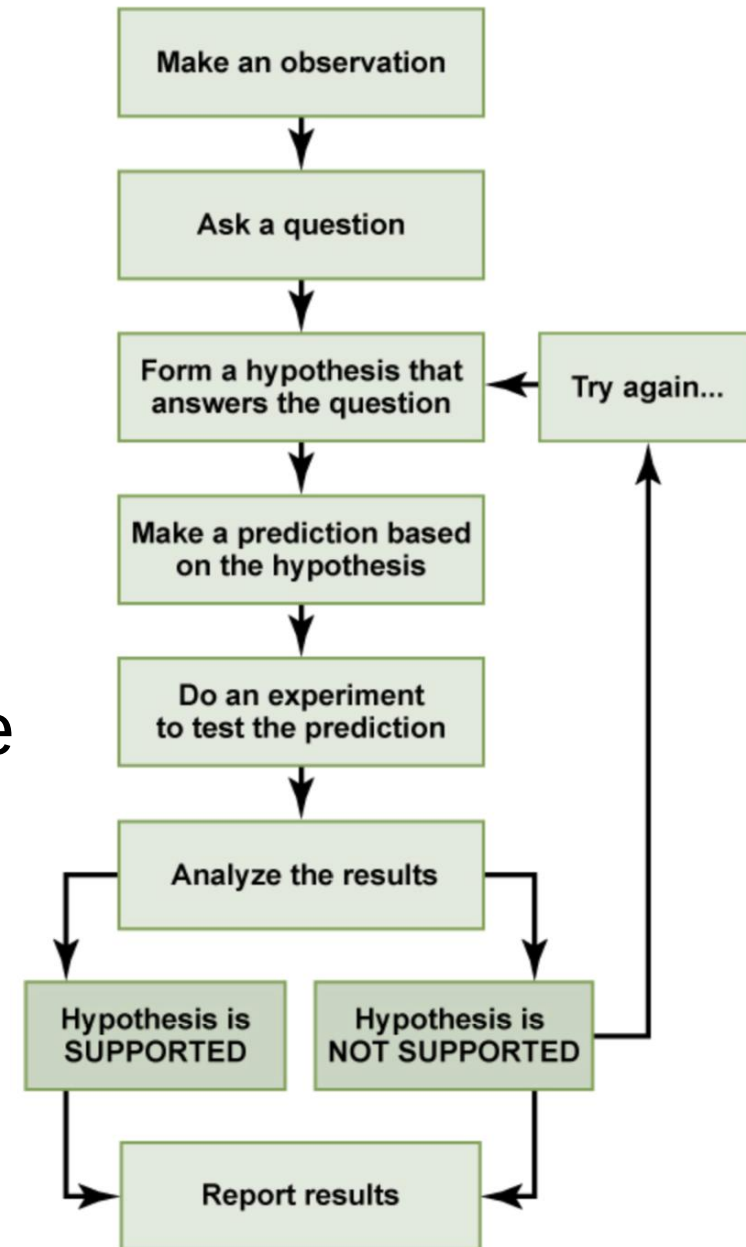
- A good detective acts carefully, going step by step to gather evidence, looking for patterns & relying on logic, evidence, & objectivity – they don't rely on a hunch or subjective biases
 - *some bad detectives do, which leads to the wrongful arrest & imprisonment of innocent people*

Copyright: Ben Phillips ben@bphillips.co.uk

- A good scientist does the same, following the **scientific method** to form a logical hypothesis & collect data/evidence in an *objective* way



- The scientific method includes:
 - A logical **hypothesis**
 - = a tentative explanation for observations
 - must be testable

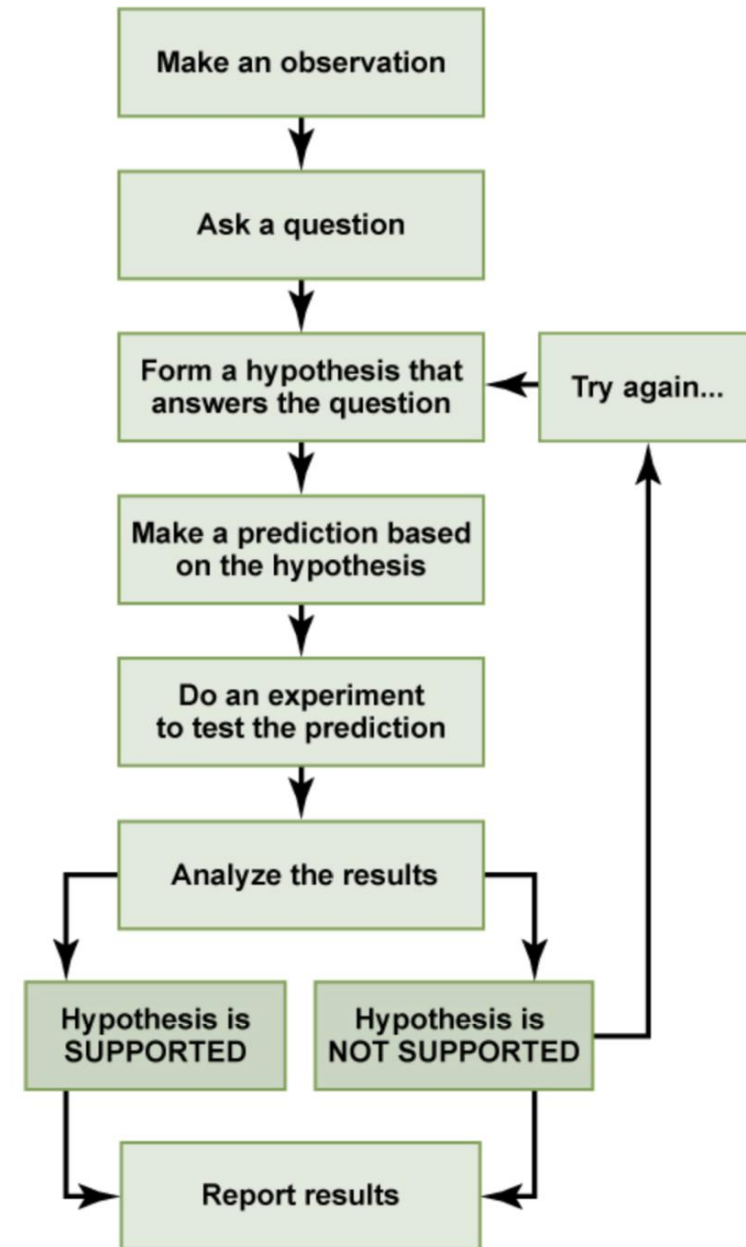


- The scientific method includes:

- A **prediction** based on the hypothesis

= if / then statement

*-If my hypothesis
is true, **then** _____
will happen*

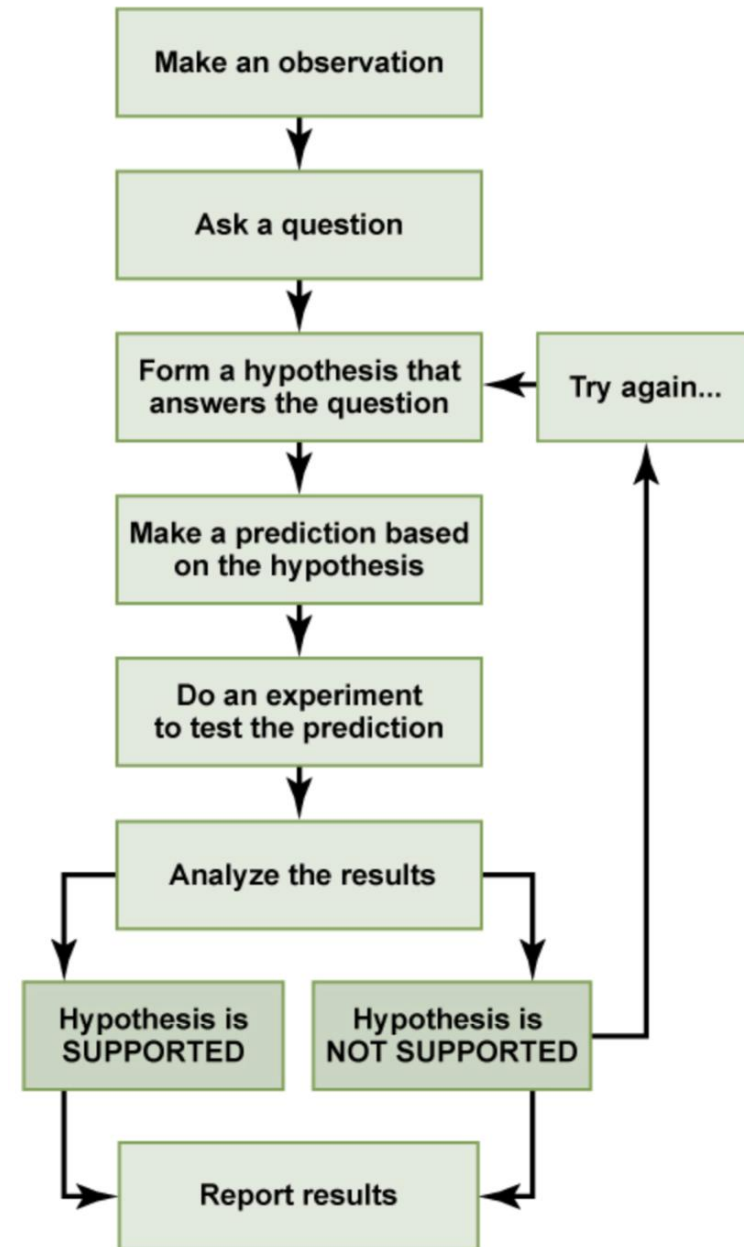


- Hypothesis & prediction examples:
 - Suppose you hypothesized that your car won't start because the battery is dead
 - Hypothesis: My car won't start because the battery is dead.
 - Prediction: If my hypothesis is correct, then my car *will* start if I put in a new battery.
 - *Another possible prediction: If my hypothesis is correct, then a different car won't start if I put my battery in it*

- The scientific method includes:

- An **experiment** or **observation** that allows you to collect data

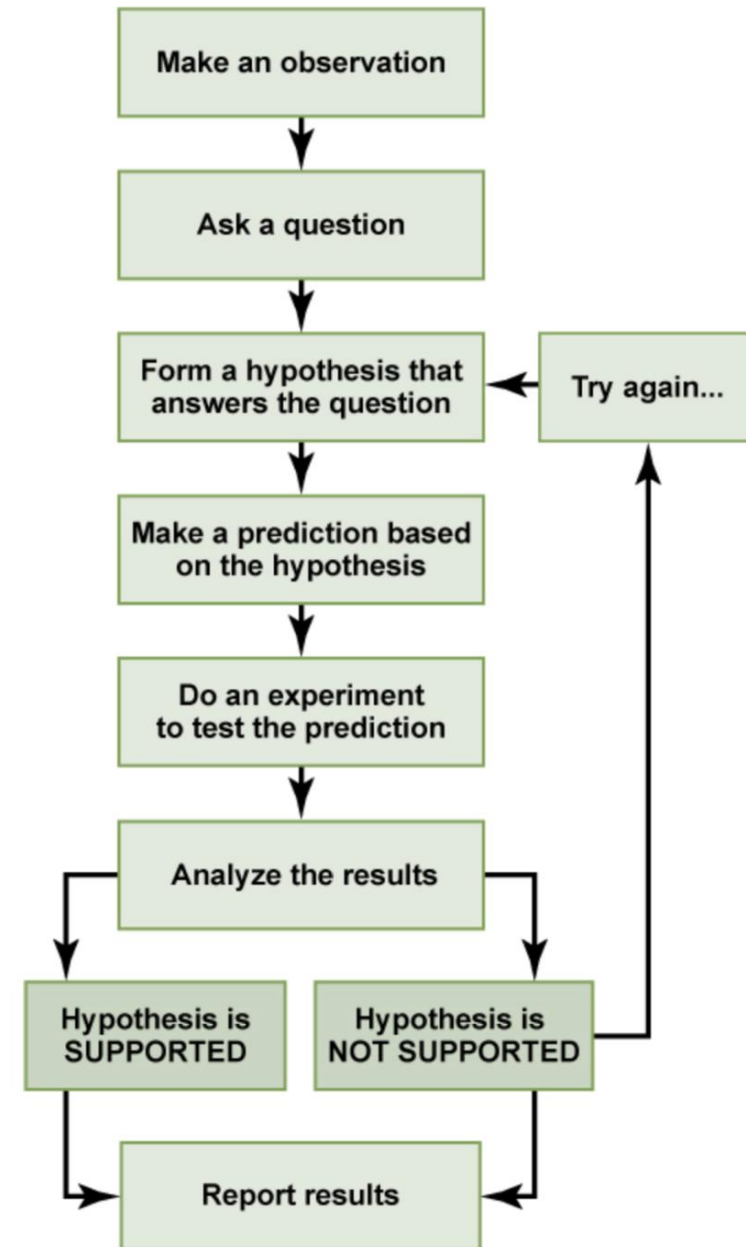
-More on setting up experiments soon



- The scientific method includes:

- Drawing a **conclusion**

-Is your hypothesis supported or not?



■ Scientific method in action:

- Hypothesis: My car won't start because the battery is dead.
- Prediction: If my hypothesis is correct, then my car *will* start if I put in a new battery.
- Experiment: Put in a new battery & try to start car.
- Conclusion:
 - If the car starts – hypothesis supported
 - If the car does not start – hypothesis rejected, come up with new hypothesis
 - e.g. My car won't start because it is out of gas.

■ Scientific method in action:

observation

My toaster doesn't toast my bread.

question

Why doesn't my toaster work?

hypothesis

There is something wrong with the electrical outlet.

prediction

If something is wrong with the outlet, then my coffeemaker also won't work when plugged into it.

experiment

I plug my coffee maker into the outlet

result

My coffeemaker works.

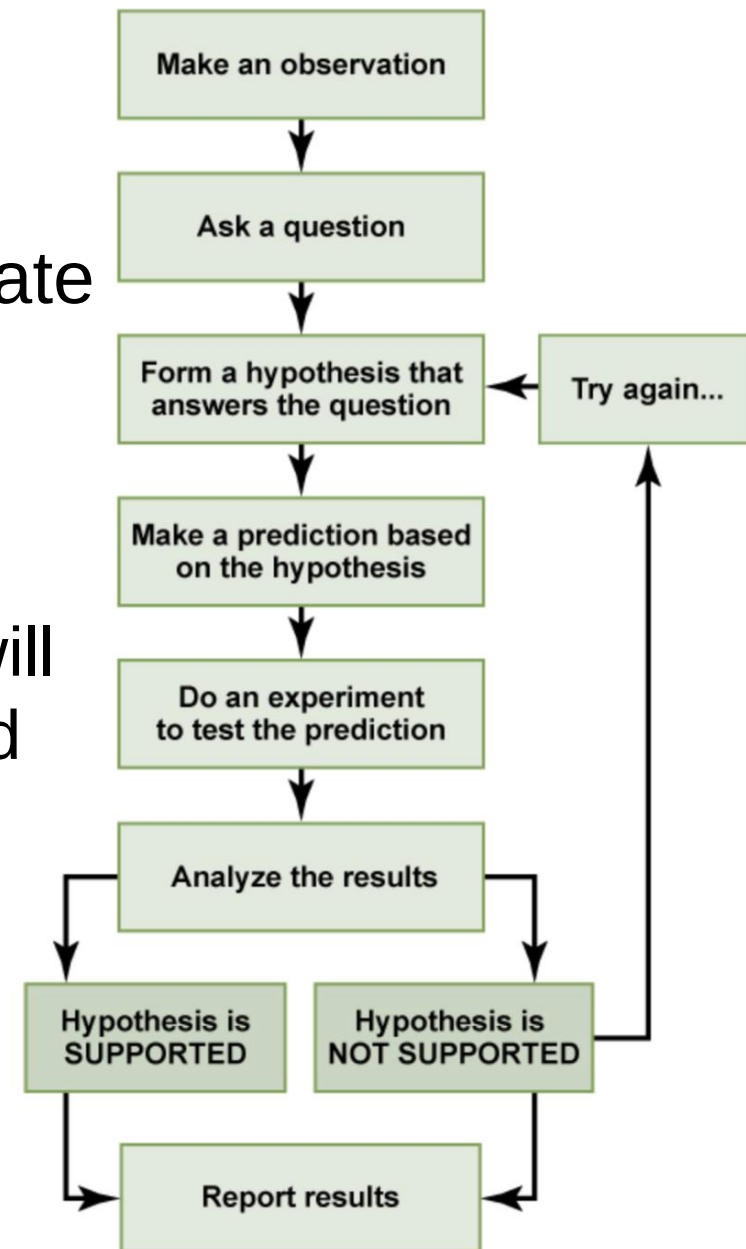
Is your hypothesis supported or not here?

■ What is **peer review**?

- Before publishing/reporting results, other scientists evaluate the validity of the methods, data, & conclusions

If they are not appropriate & well done, the conclusions will not be accepted or published

Not all journals require peer review these days, which makes it difficult for non-scientists to know what results to trust



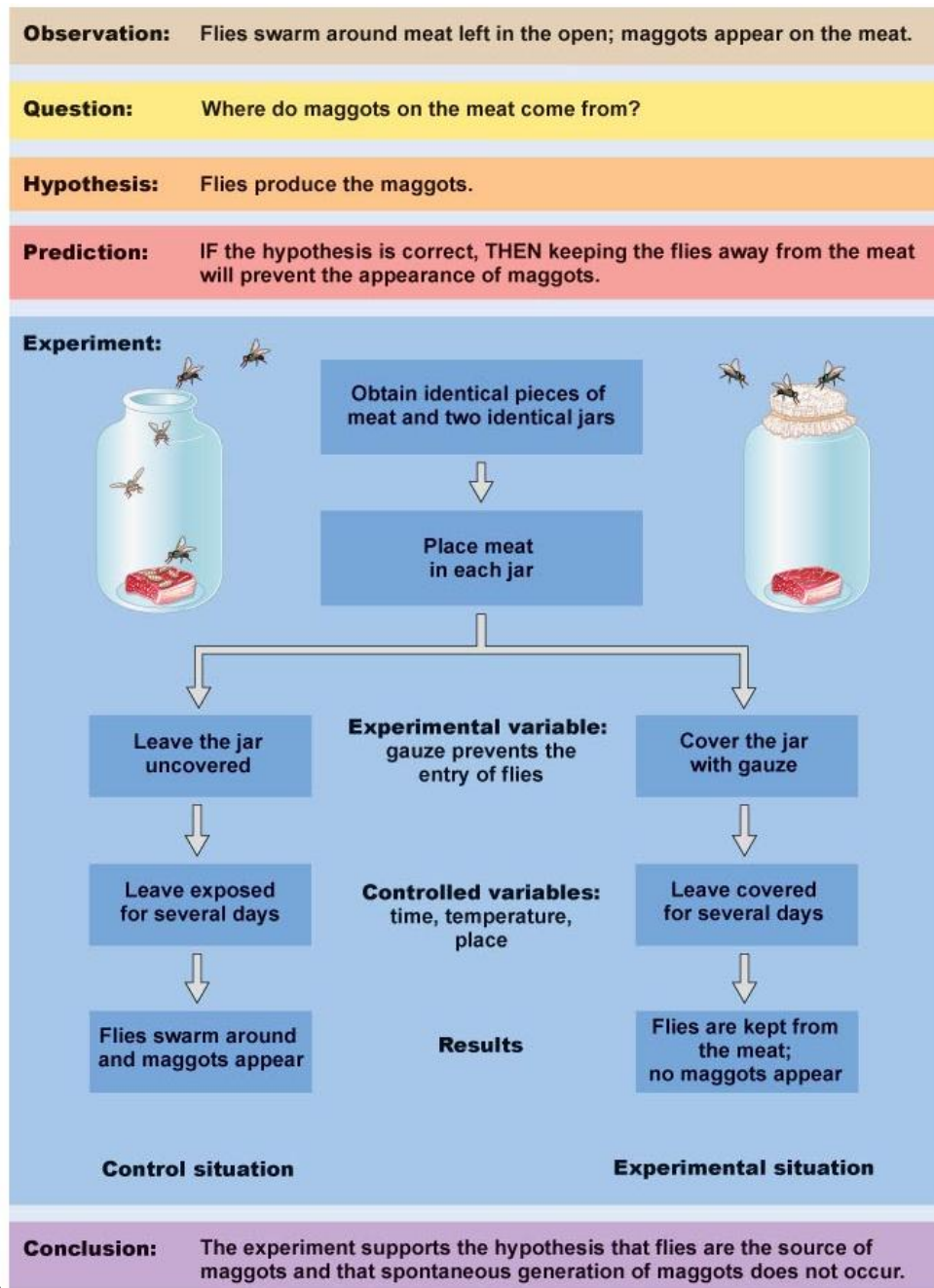
- How do we set up a good experiment?
 - We must test only ONE **variable** at a time
 - variable = changeable characteristics of an experiment
 - testing more than one at a time = **confounds**
 - We use **controls** to keep all variables (except one) the same between experimental groups
 - this prevents confounds

- How do we set up a good experiment?

Take note of:

- *the experimental group vs. the “control” group*
- *the ONE variable being tested*
- *the many variables that are “controlled” for & thus eliminated as confounds*

Figure from *Biology: Life on Earth*
Copyright: 2014 Pearson Education, Inc



The Experiment of Malte Andersson

- *Again note the controls & variables*

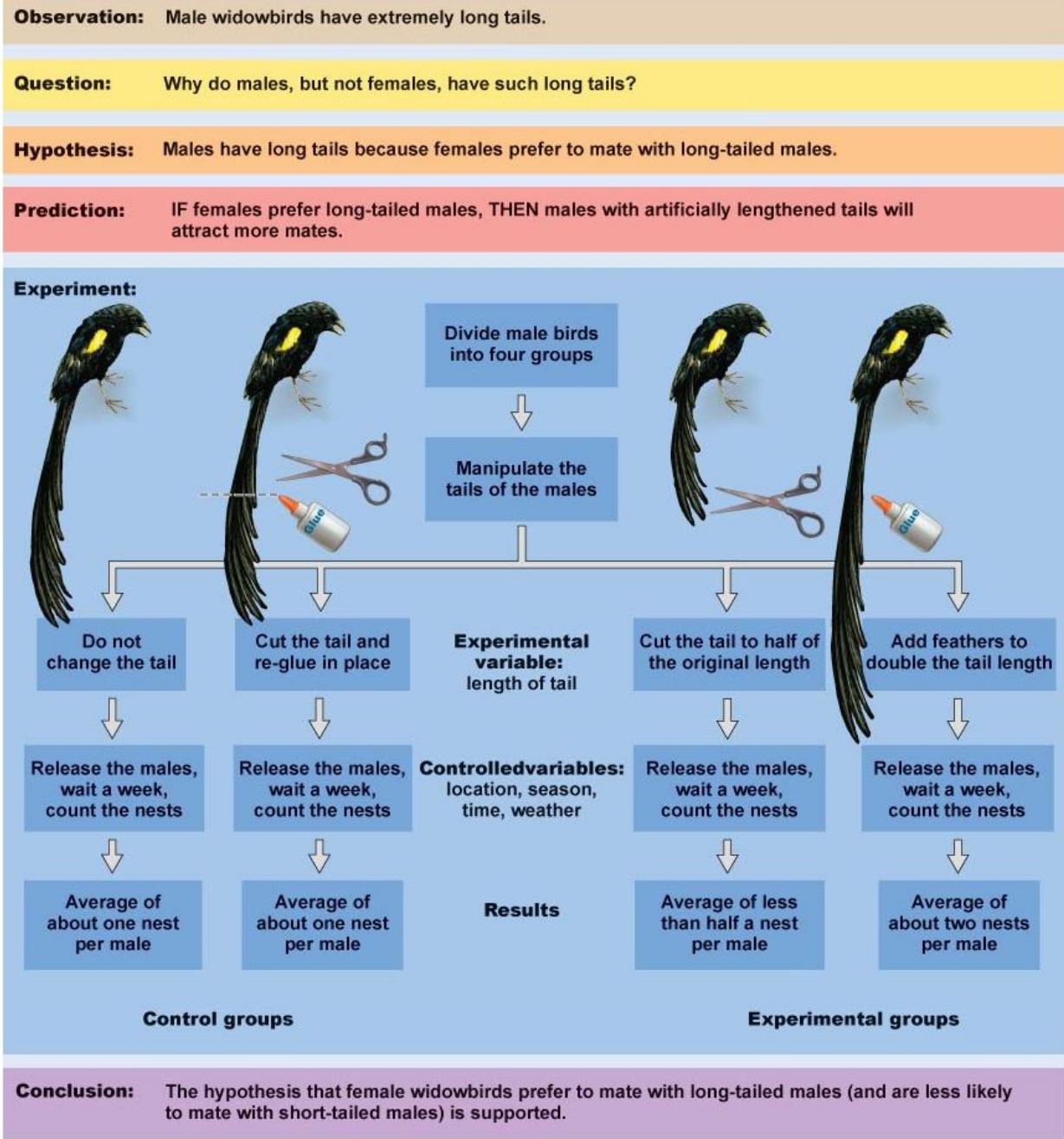


Figure from Biology: Life on Earth
Copyright: 2014 Pearson Education, Inc

■ The scientific method in action with controls:

OBSERVATION

Consuming echinacea seems to reduce the intensity or duration of symptoms of the common cold.



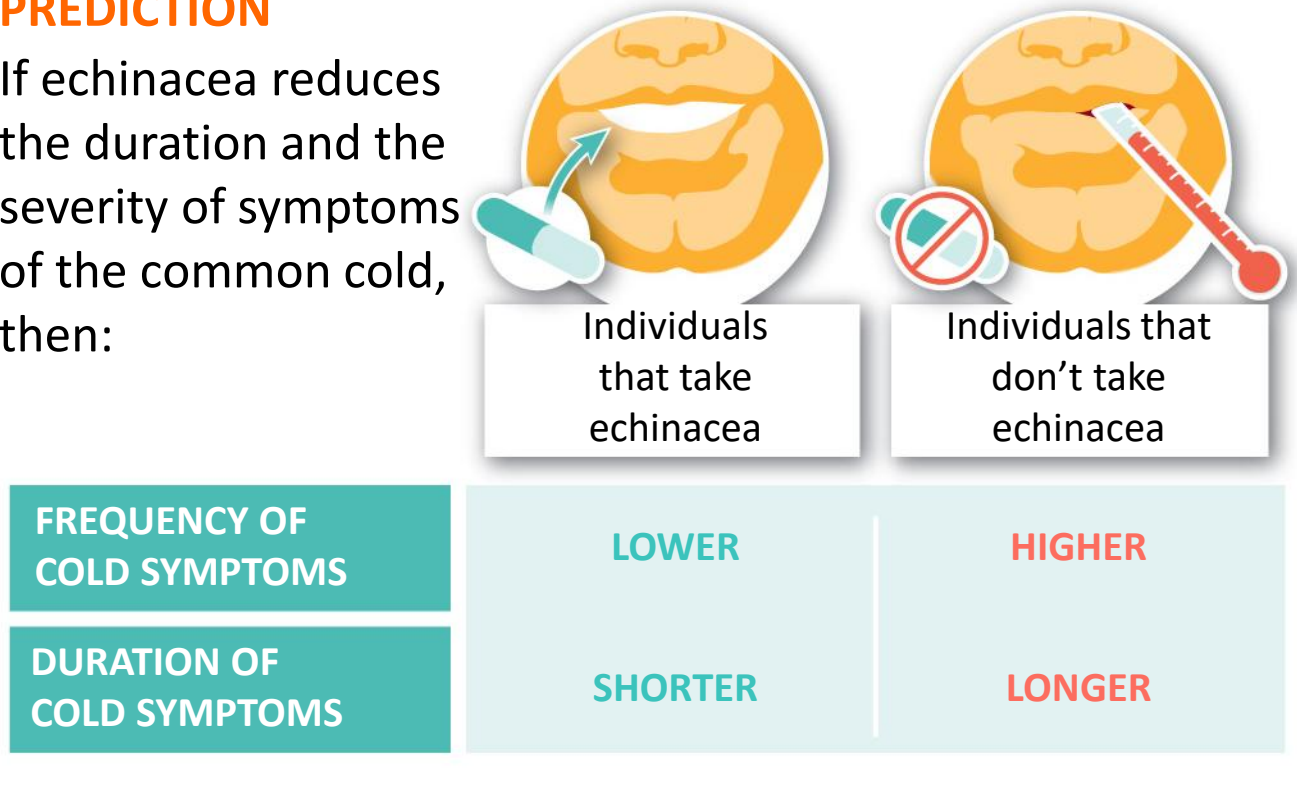
HYPOTHESIS

Echinacea reduces the duration and severity of the common cold.



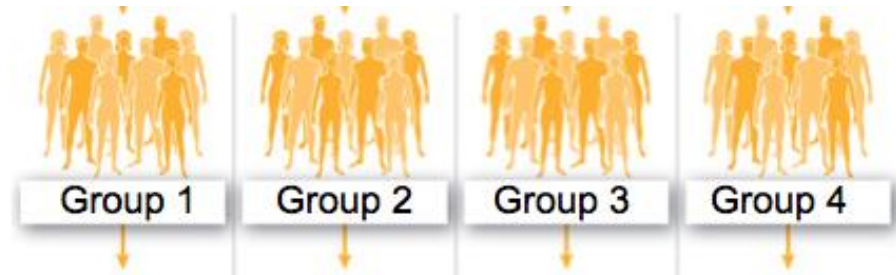
PREDICTION

If echinacea reduces the duration and the severity of symptoms of the common cold, then:



EXPERIMENT

Researchers randomly divided 437 volunteers into **four** groups to test the effect of echinacea on the common cold.



Placebo = a fake pill or substance, no active ingredient

Take note of:

- *each group and its purpose in the study*
- *the variable being tested (echinacea)*
- *the many variables that are “controlled” for and thus eliminated as confounds*
 - *e.g. time, placebo pill, same virus, etc.*

EXPERIMENT

TREATMENT (BEFORE EXPOSURE)

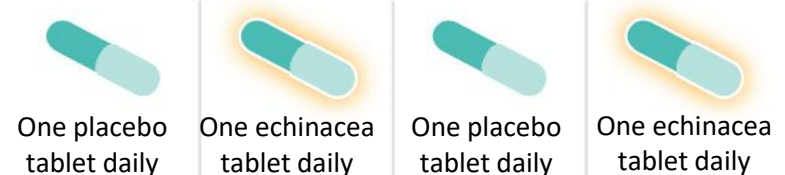
In order to determine if treatment prior to exposure has any effect on the development or duration of symptoms, groups 1 and 2 receive tablets for seven days prior to exposure, while groups 3 and 4 do not.



After one week, all individuals are exposed to a cold-causing virus.



TREATMENT (AFTER EXPOSURE)



HEALTH EVALUATION

For five days, doctors monitor all groups for cold symptoms.

RESULTS

GROUP	TREATMENT	COLD SYMPTOMS PRESENT?	DURATION OF COLD SYMPTOMS (DAYS)				
 Group 1	Placebo before and after exposure to cold virus	✓	✓ 1	✓ 2	✓ 3	○ 4	○ 5
 Group 2	Echinacea before and after exposure to cold virus	✓	✓ 1	✓ 2	✓ 3	○ 4	○ 5
 Group 3	Placebo after exposure to cold virus	✓	✓ 1	✓ 2	✓ 3	○ 4	○ 5
 Group 4	Echinacea after exposure to cold virus	✓	✓ 1	✓ 2	✓ 3	○ 4	○ 5

CONCLUSIONS

- Individuals from all four groups are equally likely to develop a cold.
- Cold symptoms lasted for the same amount of time in all groups.
- Echinacea had no effect on the duration or severity of the cold.

FURTHER EXPERIMENTATION

Alter the **amount** of echinacea given to subjects
or the **length of time** subjects receive the echinacea treatment

- In science, **repeatability** is essential
 - Once is not enough: experiments must be repeated several times for their conclusions to be valid & widely accepted
 - this is because larger, better controlled follow-up studies may undermine earlier conclusions

A small study in 2005 suggested an epilepsy drug might cure AIDS, getting people very excited



But a huge study was done & published in 2008, showing that it did not actually work

- On the next slide, you'll see two studies published only a few months apart – one suggested there is a correlation between brain structure & using marijuana, while the other suggested there is no correlation

Long-term effects of marijuana use on the brain.

Filbey FM¹, Aslan S², Calhoun VD³, Spence JS⁴, Damaraju E⁵, Caprihan A⁵, Segall J⁵.

+ Author information

<http://www.ncbi.nlm.nih.gov/pubmed/25385625>

Abstract

Questions surrounding the effects of chronic marijuana use on brain structure continue to increase. To date, however, findings remain inconclusive. In this comprehensive study that aimed to characterize brain alterations associated with chronic marijuana use, we measured gray matter (GM) volume via structural MRI across the whole brain by using voxel-based morphology, synchrony among abnormal GM regions during resting state via functional connectivity MRI, and white matter integrity (i.e., structural connectivity) between the abnormal GM regions via diffusion tensor imaging in 48 marijuana users and 62 age- and sex-matched nonusing controls. The results showed that compared with controls, marijuana users had significantly less bilateral orbitofrontal gyri volume, higher functional connectivity in the orbitofrontal cortex (OFC) network, and higher structural connectivity in tracts that innervate the OFC (forceps minor) as measured by fractional anisotropy (FA). Increased OFC functional connectivity in marijuana users was associated with earlier age of onset. Lastly, a quadratic trend was observed suggesting that the FA of the forceps minor tract initially increased following regular marijuana use but decreased with protracted regular use. This pattern may indicate differential effects of initial and chronic marijuana use that may reflect complex neuroadaptive processes in response to marijuana use. Despite the observed age of onset effects, longitudinal studies are needed to determine causality of these effects.

J Neurosci 2015 Jan 14;35(4):1505-12. doi: 10.1523/JNEUROSCI.2946-14.2015.

Daily marijuana use is not associated with brain morphometric measures in adolescents or adults.

Weiland BJ¹, Thayer RE², Depue BE³, Sabbineni A², Bryan AD², Hutchison KE².

+ Author information

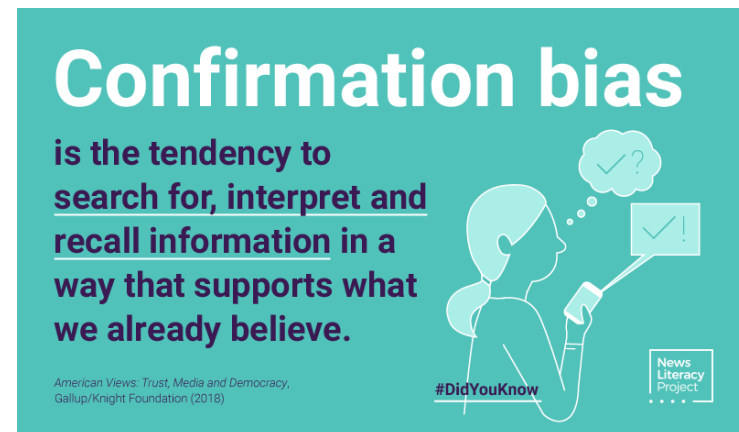
<http://www.ncbi.nlm.nih.gov/pubmed/25632127>

Abstract

Recent research has suggested that marijuana use is associated with volumetric and shape differences in subcortical structures, including the nucleus accumbens and amygdala, in a dose-dependent fashion. Replication of such results in well controlled studies is essential to clarify the effects of marijuana. To that end, this retrospective study examined brain morphology in a sample of adult daily marijuana users (n = 29) versus nonusers (n = 29) and a sample of adolescent daily users (n = 50) versus nonusers (n = 50). Groups were matched on a critical confounding variable, alcohol use, to a far greater degree than in previously published studies. We acquired high-resolution MRI scans, and investigated group differences in gray matter using voxel-based morphometry, surface-based morphometry, and shape analysis in structures suggested to be associated with marijuana use, as follows: the nucleus accumbens, amygdala, hippocampus, and cerebellum. No statistically significant differences were found between daily users and nonusers on volume or shape in the regions of interest. Effect sizes suggest that the failure to find differences was not due to a lack of statistical power, but rather was due to the lack of even a modest effect. In sum, the results indicate that, when carefully controlling for alcohol use, gender, age, and other variables, there is no association between marijuana use and standard volumetric or shape measurements of subcortical structures.

■ Things to avoid:

- **Bias** – all people, including scientists, have biases, which can be subconscious (we don't realize them), so avoiding them is a challenge
 - e.g. most journals hide author names during the peer review process because female authors are less likely to have papers accepted when the reviewers know they're female (even though most reviewers believe they are not sexist)
 - e.g. I once knew a student that would re-measure something when it didn't fit his hypothesis, but only measure once when it did – this is “confirmation bias”



■ Things to avoid:

– **Pseudoscience** = practices that claim to be factual but incompatible with the scientific method

- often have exaggerated or unfalsifiable claims &/or no peer review

Anti-vaccination websites use 'science' and stories to support claims, study finds

Anti-vaccine positions are frequently embedded with information about positive behaviors like eating healthy, breastfeeding

Date: November 3, 2015

Source: Johns Hopkins University Bloomberg School of Public Health

Summary: A content analysis of nearly 500 anti-vaccination websites found that over two-thirds used what they represented as scientific evidence to support the idea that vaccines are dangerous and nearly one-third contained anecdotes that reinforced the perception.

FULL STORY

A content analysis of nearly 500 anti-vaccination websites found that over two-thirds used what they represented as scientific evidence to support the idea that vaccines are dangerous and nearly one-third contained anecdotes that reinforced the perception.

→ The sites contained a considerable amount of misinformation and pseudoscience, with more than two-thirds suggesting that vaccines were dangerous, just under two-thirds suggesting they cause autism and just over 4 in 10 claiming vaccines cause "brain injury." More than two-thirds used what they represented as scientific evidence that in fact was not, while about 3 in 10 used anecdotes to support these claims.

<https://www.sciencedaily.com/releases/2015/11/151103134800.htm>

– **Relying on anecdotes**

- these are subject to extreme cognitive bias & are cherry-picked – they often represent the exception, not the rule

- Things to avoid:
 - **People that want your money**

- Example 1: for 15+ years Airborne stated it boosts your immune system, based on a “rigorous clinical study”
- A class-action lawsuit revealed there was no study ever done – it was completely made-up!
- They had to remove the false claim but sales haven’t dropped yet consumers aren’t aware

The truth: Airborne does not reduce how often you get sick or how long you are sick

★ THIS STATEMENT HAS NOT BEEN EVALUATED BY THE FOOD AND DRUG ADMINISTRATION. THIS PRODUCT IS NOT INTENDED TO DIAGNOSE, TREAT, CURE OR PREVENT ANY DISEASE.



photo by Mike Mozart

- People that want your money example 2:
 - In 1998 Andrew Wakefield published data that MMR vaccines (measles, mumps, rubella) caused autism
 - it was found later that he made up the results completely – the results were fake!
 - Why would he fake the data?
 - Wakefield had patented a new vaccine & wanted people to buy his instead
 - parents of children with autism payed him for the research so they could sue the vaccine company

by Brian Deer

Sunday February 22 2004,
12.00am GMT, The Times

Revealed: MMR research scandal

FULL details are disclosed today of the four-month Sunday Times investigation that has uncovered a medical scandal at the heart of the worldwide scare over MMR.

Andrew Wakefield, the doctor who champions the alleged link between measles, mumps and rubella vaccine and autism in young children, stands discredited for misleading his medical colleagues and The Lancet, the professional journal that published his findings.

The investigation has found that when he warned parents to avoid MMR, and published research claiming a link with autism, he did not disclose he was being funded through solicitors seeking evidence to use against vaccine manufacturers.

<https://www.thetimes.co.uk/article/revealed-mmr-research-scandal-7ncfntn8mjg>

Photo by Bladość



- Wakefield's paper was fully retracted, he lost his medical license, & lost his appointments to medical organizations
- However: the results of his paper are still evident today – some parents now decide NOT to vaccinate their children based on these fears

– Many diseases were once eradicated in the US, but in the last few years they have made a comeback due to low vaccination rates

e.g. measles, mumps, rubella, whooping cough: most can be fatal or cause birth defects



- Things to avoid:

- **Sensationalism** = reporters may be trying to alarm you to get you to click on their articles, or they may have good intentions but not enough knowledge to accurately report the findings
- Remember the study that showed a correlation between brain structure & marijuana use? It made no claims about causation or that the structural differences were bad (for all we know they were good differences), but:

Medscape Medical News > Psychiatry

Early Marijuana Use: A More Harmful Impact on Brain Structure

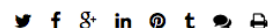
Deborah Brauser
November 11, 2014

Bad News For Long-Term Marijuana Users, Study Says

Randall Mayes , Design & Trend
Nov, 15, 2014, 02:56 PM

New Study Indicates Long-term Effects of Weed Use on Brain, Dire Implications for Pot Proponents

Michael Tremoglie | +Follow
Nov 21, 2014 4:23 PM EST



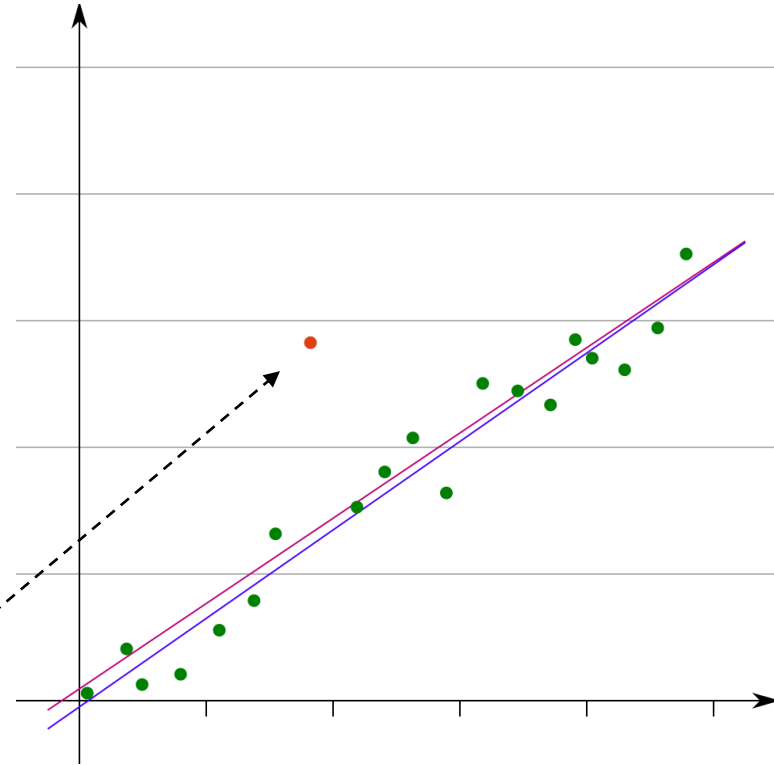
Cannabis shrinks brain? Study says pot abuse damages IQ

Published time: November 11, 2014 13:36

- Things to avoid:

- **Getting hung up on a unique study:** if a few small studies show one thing, but the vast majority of studies show something else, go with the majority

- one of the reasons repeatability is so vital in science is because a number of things can occur in a study that can throw off the data or give inaccurate data



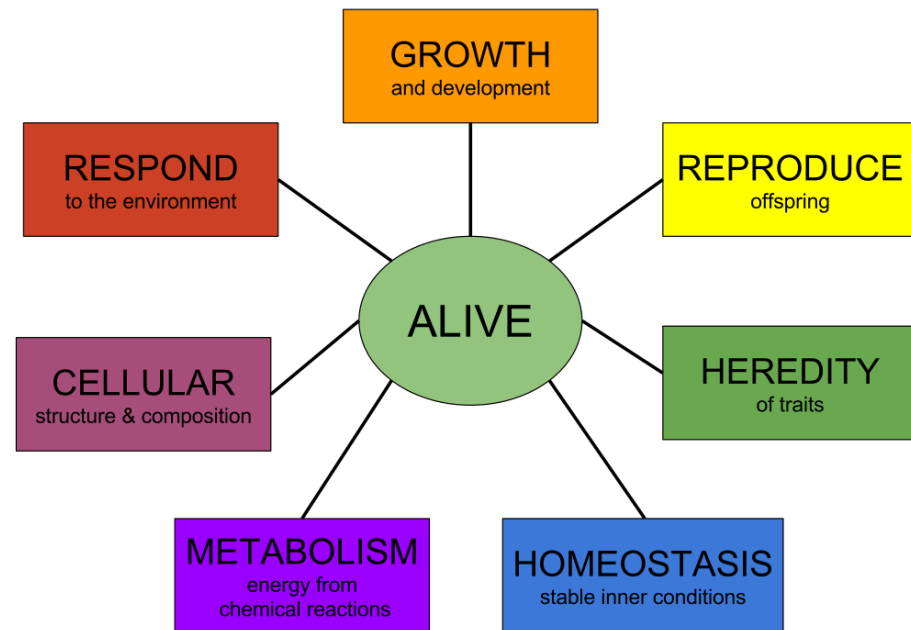
Why do scientists use the word “theory?”

- A scientific *theory* differs from everyday usage
 - Many people use the word *theory* to mean hypothesis, or an “educated guess”
- **Scientific theory** = an explanation about something in the natural world that:
 - can be extensively and repeatedly tested
 - uses facts to *explain* the natural world
 - established scientific theories have been rigorously tested and are supported by all (or the vast majority of) observations/experiments

- **Facts & laws** only *describe* things in the natural world
 - repeatable observations that everyone agrees on
 - e.g. it is a fact that a dropped pencil falls toward the ground (the “law” of gravity)
- **Theories** *explain* things in the natural world
 - e.g. the “theory” of gravity explains the forces that cause pencils and other objects to fall
 - Common established theories in biology:
 - Cell theory: all organisms are composed of cells
 - Evolution: populations change over many generations due to natural selection

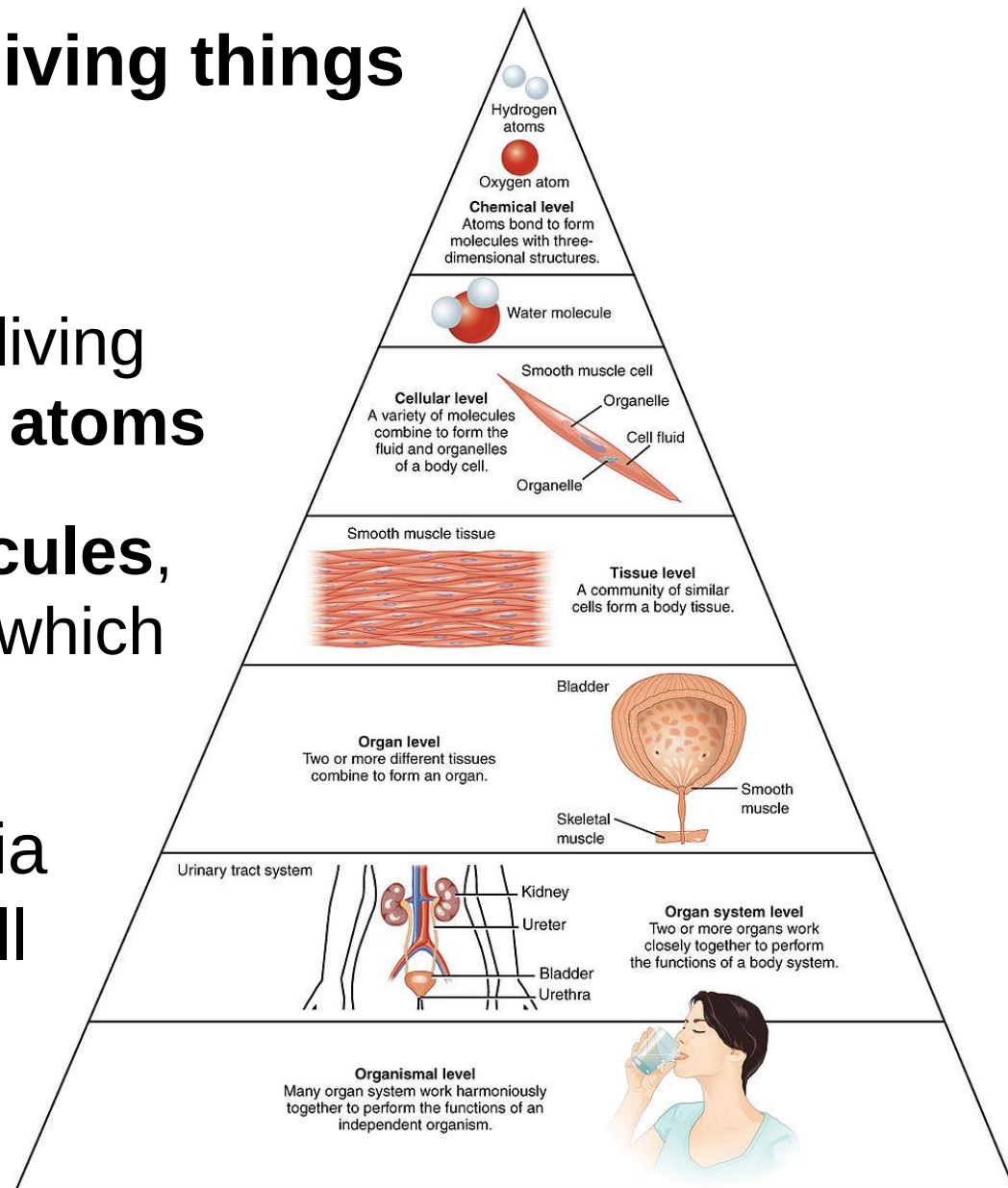
What is biology & what makes a thing “living?”

- **Biology** = the scientific study of living things
- How do we know if a thing is living?
 - Can't define life by it's parts: we can put proteins, fats, DNA, etc. in a test tube but it's not alive
 - Instead of a definition, scientists agree some characteristics of life: need all to be a living thing



■ A characteristic of life: **living things are organized**

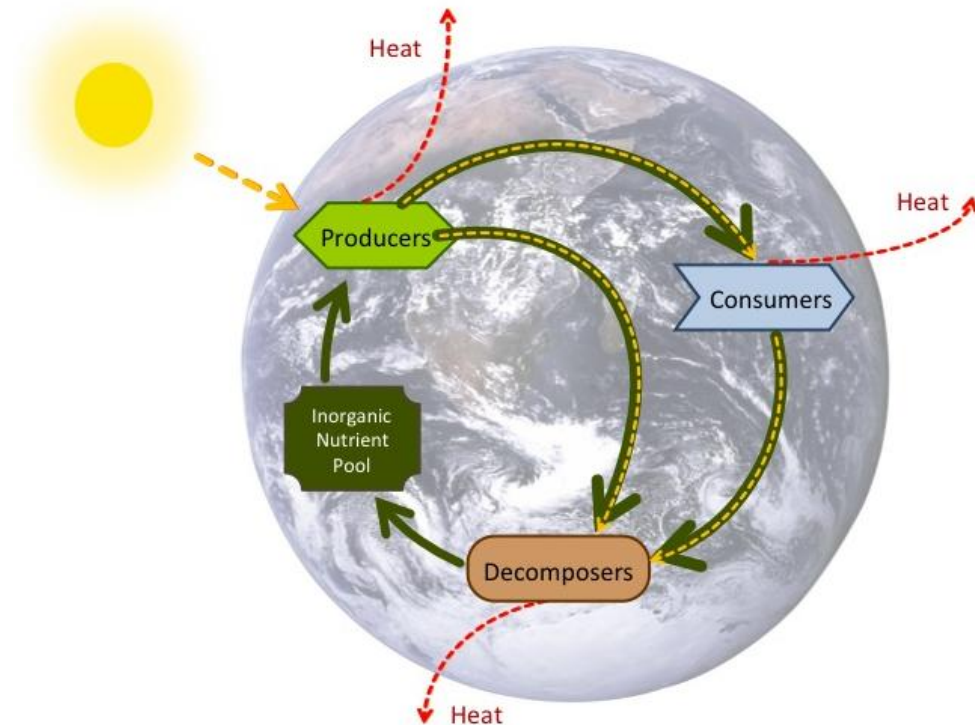
- At the smallest scale, living things are made up of **atoms**
- Atoms make up **molecules**, which make up **cells**, which make up living things
- From trees to bacteria to humans, we are all organized in the same way



- A characteristic of life: **living things require energy & nutrients**

- Energy comes from the sun or from consuming other organisms
- Nutrients come from the nonliving environment or from consuming living organisms

Dark green lines represent the movement of nutrients, and dashed lines represent the movement of energy. As you can see, nutrients remain within the system while energy enters via photosynthesis and leaves primarily as heat energy.

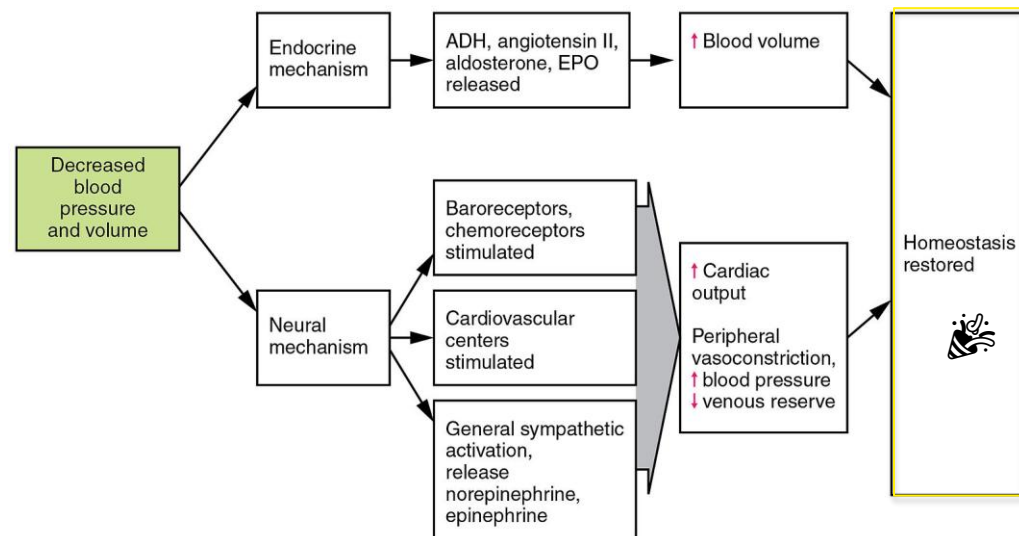


- A characteristic of life: **living things maintain homeostasis**

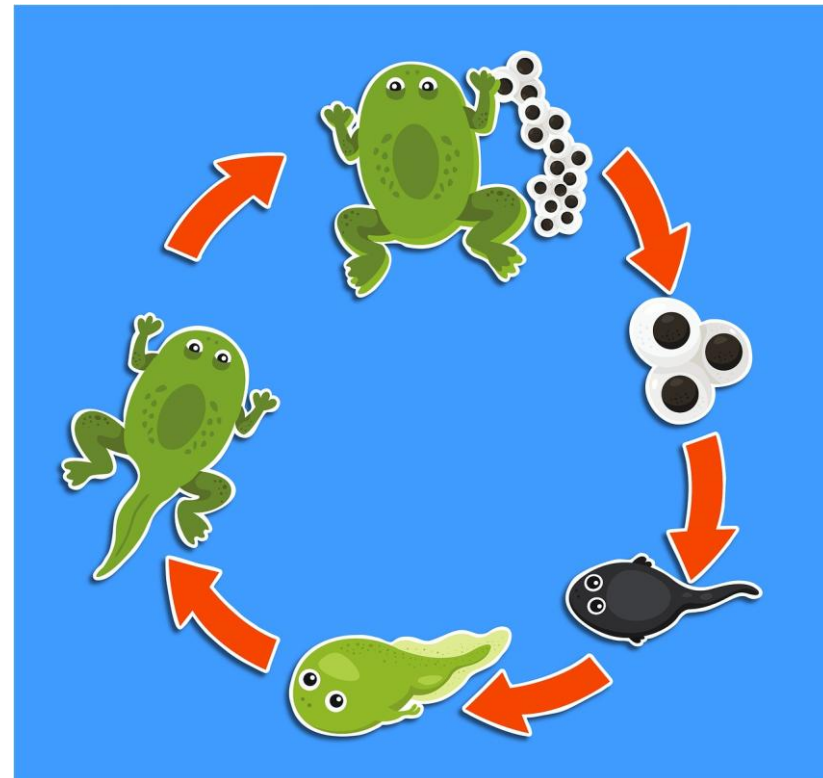
- **Homeostasis** = process by which an organism regulates their body, maintaining *equilibrium* (internal constancy) even as the surrounding environment changes

- e.g. temperature, blood pressure, pH

- *This theme is covered in almost every chapter*



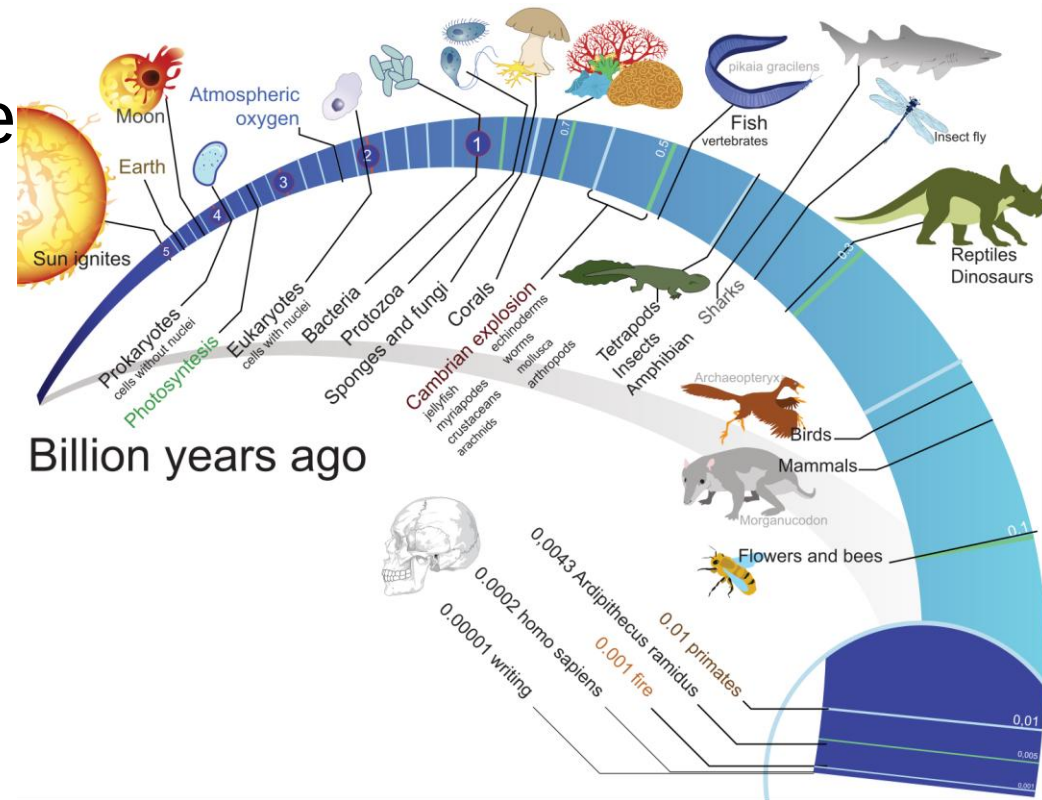
- A characteristic of life: **living things grow & develop**
 - Our DNA guides the development & expression of our traits
 - We grow from infants into adulthood



- A characteristic of life: **living things reproduce**
 - Reproduction is the transmission of DNA from generation to generation
 - this must happen at a species level, not necessarily at the individual level (i.e. you don't have to individually reproduce to be alive, but at least some members of your species must)



- A characteristic of life: **living things evolve**
 - **Evolution** = change in genetic make-up of a population over multiple generations
 - does not occur in individuals, but instead takes time over many generations



- Characteristic of life: **living things evolve**

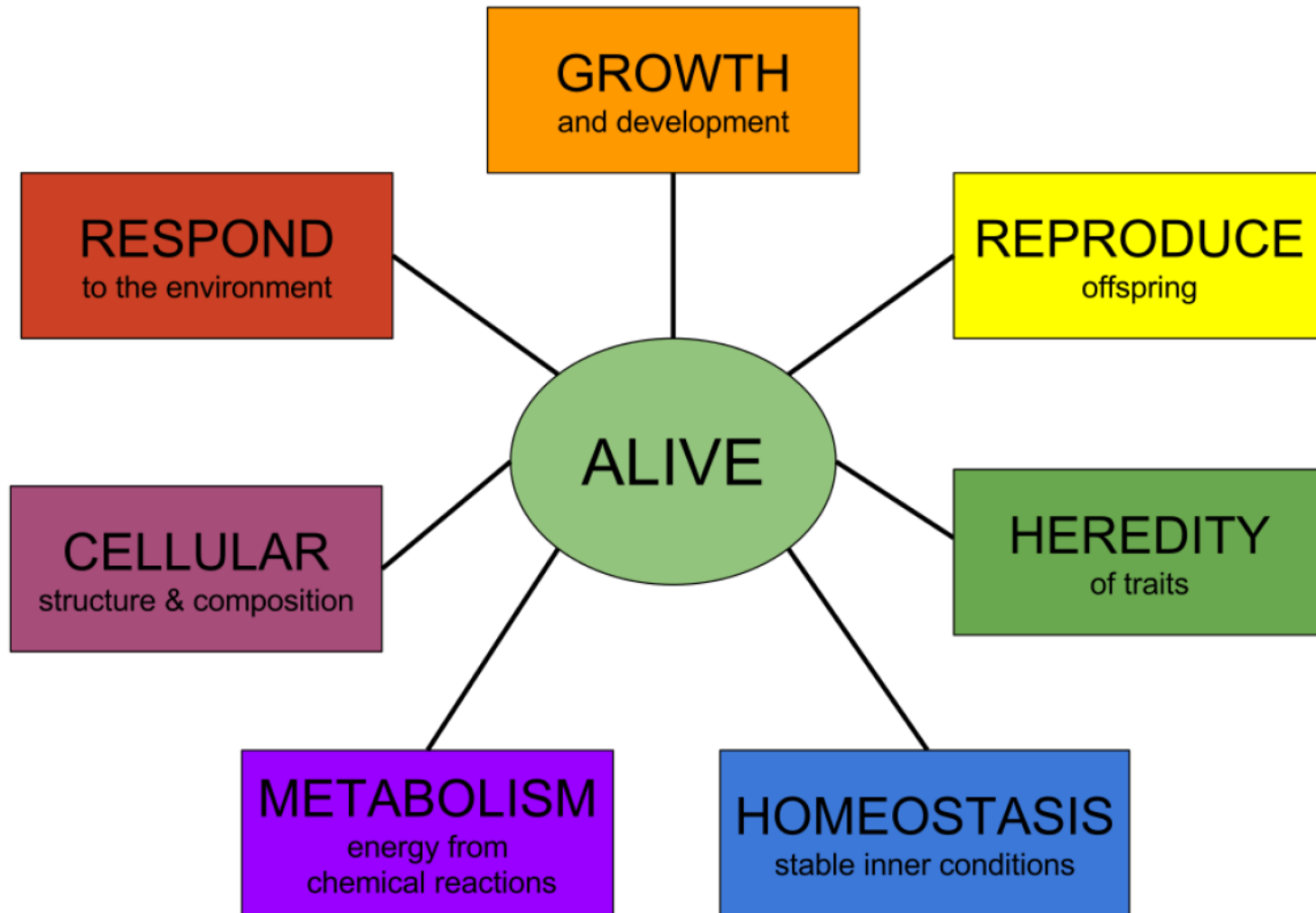


Figure: <http://commons.wikimedia.org/wiki/User:Earthdirt>

Content covered today

LHS Objectives

LHS Importance

Scientific Study

Biology